AP® BIOLOGY · UNIT 2

Cell Structure and Function

The Comprehensive Study Guide
Organelles, Membranes, Transport, and Compartmentalization

Step-by-step lessons for all eleven Unit 2 topics
3D organelle gallery · Labeled diagrams · Misconception alerts
Worked examples: surface area-to-volume and water potential
3 one-page mind maps · 83 exam-style multiple-choice questions
100 “Who Am I?” vocabulary questions
Free-response practice with scoring guidelines
Full answer explanations · One-page review sheet

DAVID SREERAMA

AP® Biology Unit 2: Cell Structure and Function — The Comprehensive Study Guide

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Illustrations. Cell and organelle illustrations are simplified models drawn to show structure and relationships. They are not to scale.

Disclaimer. This guide is intended as a supplementary study aid. Course content and exam formats can change; always check the current AP Biology Course and Exam Description published by the College Board for the most up-to-date information. While every effort has been made to ensure accuracy, the author and publisher assume no responsibility for errors or omissions.

First edition

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How to Use This Guide

Unit 2 moves biology from molecules to the cell. The chemistry of Unit 1—polarity, hydrogen bonds, hydrophobic interactions, and the structure of proteins and phospholipids—now explains how real cells are built and how they control what goes in and out. Students who can connect a *cell structure* to the *job it does* find Units 3 through 6 much easier, because respiration, photosynthesis, cell signaling, and gene expression all happen in the organelles and membranes you meet here.

This guide covers all eleven topics of AP Biology Unit 2. Each topic follows the same pattern so you always know where to look:

- **Learning Targets** — what you should be able to do by the end of the lesson. Use them as a checklist.
- **The Lesson** — a clear, step-by-step explanation with 3D illustrations, diagrams, and tables.
- **Worked Examples** — the Unit 2 calculations (surface area-to-volume ratio and water potential) solved one step at a time.
- **Misconception Alerts** — the mistakes that cost students points year after year.
- **Exam Tips** — how the idea tends to be tested and how to phrase strong answers.
- **Key Terms** — the vocabulary you must be able to use precisely.
- **Check Your Understanding** — exam-style multiple-choice questions. Full explanations are in Part 4.

Topic 2.1 opens with a **3D Organelle Gallery**: one card per structure showing what it looks like, where it is found, and what it does. After the lessons, three **mind maps** summarize the unit on single pages, and the **Who Am I? Vocabulary Challenge** in Part 3 gives you 100 quick clues to sharpen your vocabulary.

A Four-Week Study Plan

WEEK	WHAT TO DO
Week 1	Read Topics 2.1 and 2.2. Study the Organelle Gallery until you can name each organelle's function from its picture. Trace the path of a secreted protein from ribosome to plasma membrane from memory.
Week 2	Read Topics 2.3, 2.4, and 2.5. Work every surface area-to-volume example by hand. Draw and label the fluid mosaic model from memory.
Week 3	Read Topics 2.6 through 2.9. Make a table of every transport mechanism (energy? protein? direction?). Practice the water potential examples until you can do them without looking.
Week 4	Read Topics 2.10 and 2.11. Study the mind maps, the Master Comparisons, and the One-Page Review Sheet. Work through the Who Am I? challenge. Take the full Practice Test under timed conditions, score it, and record your results in the Score Tracker.

Study Habits That Work

- **Retrieve, don't reread.** Close the book and list every organelle and its function, then check. Testing yourself is far more effective than highlighting.
- **Draw it.** If you can sketch an animal cell, a plant cell, the fluid mosaic membrane, and cells in hypotonic and hypertonic solutions from memory, you understand Unit 2.

- **Always name the direction.** For every transport question, say which way the substance moves relative to its concentration gradient and whether energy is used.
- **Explain the "because."** AP questions rarely ask *what* an organelle is; they ask *why* its structure suits its function and *what would happen if* it failed.
- **Space it out.** Return to earlier topics every few days. Short, repeated reviews beat one long cram session.

Unit 2 at a Glance

On the AP Biology Exam, Unit 2 has historically made up roughly **10–13% of the multiple-choice score**. Its ideas—membranes, gradients, compartments—are the foundation for cellular energetics (Unit 3) and cell communication (Unit 4), so its true weight on the exam is larger than that number suggests.

TOPIC	TITLE	THE BIG QUESTION
2.1	Cell Structure: Subcellular Components	What are the parts of a cell, and which cells have them?
2.2	Cell Structure and Function	How does the structure of each organelle let it do its job?
2.3	Cell Size	Why are cells small, and how does surface area-to-volume ratio limit them?
2.4	Plasma Membranes	How is the plasma membrane built, and why is it described as a fluid mosaic?
2.5	Membrane Permeability	Which substances can cross the membrane on their own, and which cannot?
2.6	Membrane Transport	How do passive and active transport differ?
2.7	Facilitated Diffusion	How do channel and carrier proteins move polar molecules and ions?
2.8	Tonicity and Osmoregulation	Which way will water move, and how do organisms control it?
2.9	Mechanisms of Transport	How do pumps, cotransporters, and vesicles move substances?
2.10	Cell Compartmentalization	Why do eukaryotic cells divide their interior into compartments?
2.11	Origins of Cell Compartmentalization	Where did mitochondria and chloroplasts come from?

THE ONE BIG IDEA

Membranes create boundaries, and boundaries make life possible. The plasma membrane separates the cell from its surroundings so the cell can keep a different internal environment. Internal membranes divide eukaryotic cells into compartments where different reactions can run at the same time without interfering. Because membranes are selectively permeable, cells can build up *concentration gradients*—and gradients store energy that cells use to make ATP, send signals, and move substances. On almost every Unit 2 free-response question, the winning answer connects a *specific structure* (a membrane, a protein, a fold, a compartment) to a *specific function*.

Connections to Other Units

UNIT 2 IDEA	WHERE YOU WILL SEE IT AGAIN
Phospholipid bilayer and amphipathic molecules	Unit 1 explained *why* phospholipids form bilayers; Unit 2 shows what the bilayer does.
Folded inner membranes (cristae, thylakoids)	Unit 3 — Electron transport chains and chemiosmosis in respiration and photosynthesis.
Proton and ion gradients	Unit 3 — ATP synthase. Unit 4 — Nerve signals and membrane potential.

UNIT 2 IDEA	WHERE YOU WILL SEE IT AGAIN
Membrane receptors and glycoproteins	Unit 4 — Cell signaling and cell recognition.
Ribosomes, rough ER, Golgi	Unit 6 — Translation and protein targeting.
Endosymbiosis and shared features of cells	Unit 7 — Common ancestry and the evolution of eukaryotes.

PART 1 · THE LESSONS

2.1

TOPIC 2.1

Cell Structure: Subcellular Components

LEARNING TARGETS

- Describe the structure and function of the ribosome, endoplasmic reticulum, Golgi complex, mitochondria, lysosomes, vacuoles, and chloroplasts.
- Explain why ribosomes in every living cell are evidence of common ancestry.
- Compare prokaryotic and eukaryotic cells, and plant and animal cells.
- Identify organelles in a diagram or micrograph and state their function.

Step 1: What Every Cell Has

The **cell theory** states that all living things are made of one or more cells, that the cell is the basic unit of life, and that all cells come from pre-existing cells. Despite the enormous diversity of life, **every cell**—bacterium, amoeba, leaf cell, or neuron—has four things in common:

1. **A plasma membrane** — a phospholipid bilayer that separates the inside of the cell from the outside and controls what crosses (Topics 2.4–2.9).
2. **Cytoplasm** — everything inside the membrane. The fluid part is the **cytosol**, a water-based solution where many reactions occur.
3. **DNA** — the genetic material, which carries instructions for building proteins.
4. **Ribosomes** — the structures that build proteins.

Step 2: Prokaryotic vs. Eukaryotic Cells

Cells are divided into two basic types. **Prokaryotic cells** (bacteria and archaea) lack a nucleus and membrane-bound organelles; their DNA sits in a region of the cytoplasm called the **nucleoid**. **Eukaryotic cells** (protists, fungi, plants, and animals) have a true nucleus surrounded by a double membrane, plus many internal membrane-bound organelles.

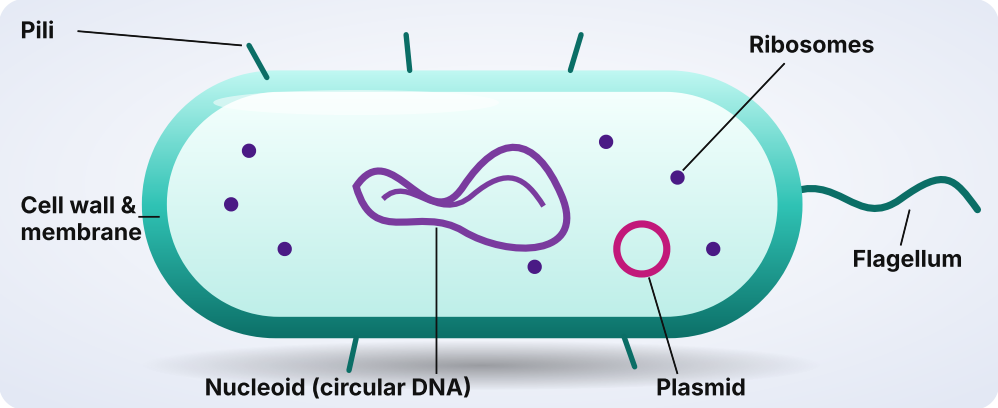


Figure 2.1 A prokaryotic (bacterial) cell. There is no nucleus and no membrane-bound organelles. The single circular chromosome lies in the nucleoid; small extra DNA circles called plasmids may carry genes such as antibiotic resistance. Ribosomes are present, but they are smaller than eukaryotic ribosomes.

FEATURE	PROKARYOTIC CELL	EUKARYOTIC CELL
Examples	Bacteria, archaea	Protists, fungi, plants, animals
Nucleus	None; DNA in the nucleoid	DNA enclosed in a nucleus with a double membrane
DNA	Usually one circular chromosome; may have plasmids	Multiple linear chromosomes
Membrane-bound organelles	None	Many: ER, Golgi, mitochondria, lysosomes, and more
Ribosomes	Yes (smaller)	Yes (larger)
Size	Usually 1–5 µm	Usually 10–100 µm
Cell wall	Usually present (peptidoglycan in bacteria)	Plants (cellulose) and fungi (chitin); none in animals

◆ **MISCONCEPTION ALERT**

"Prokaryotes have no organelles" is not quite right. Prokaryotes have no *membrane-bound* organelles, but they do have ribosomes, which are organelles without a membrane. Some prokaryotes also have internal folds of the plasma membrane where reactions such as photosynthesis take place.

Step 3: Tour the Eukaryotic Cell

The figures below show a typical animal cell and a typical plant cell cut open so you can see inside. Study them before reading the Organelle Gallery, then come back and test yourself by covering the labels.

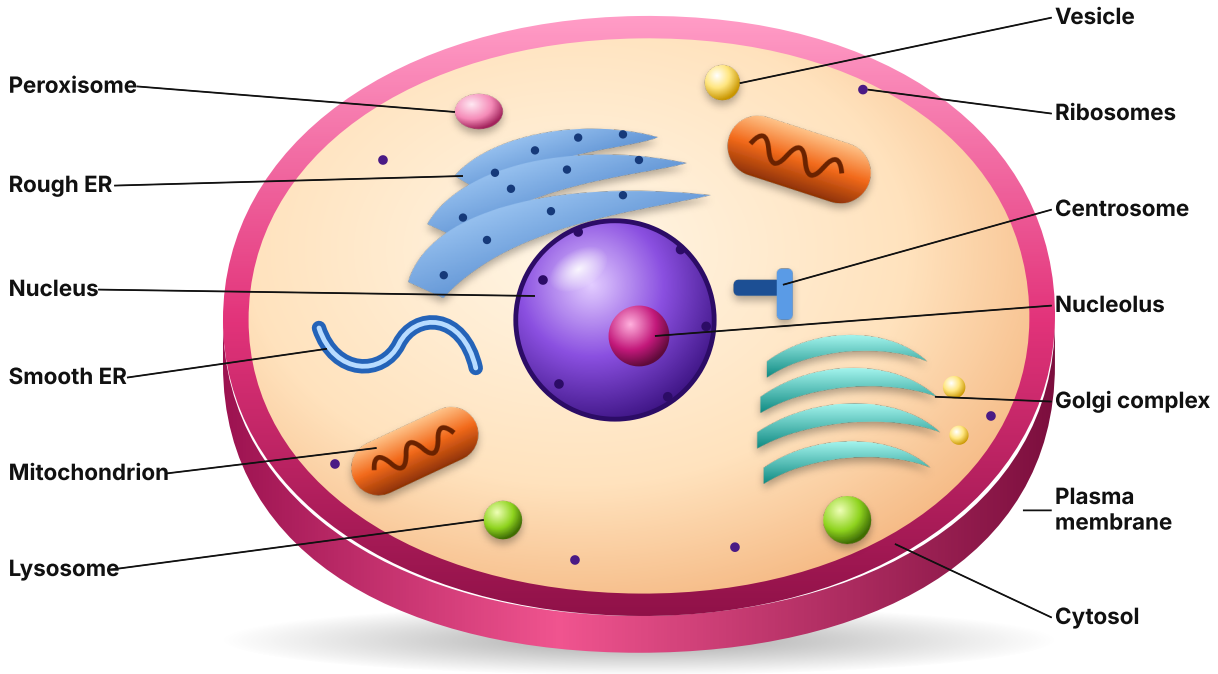


Figure 2.2 A 3D cutaway of an animal cell. Note what is *absent*: animal cells have no cell wall, no chloroplasts, and no large central vacuole. They do have centrosomes with centrioles and lysosomes.

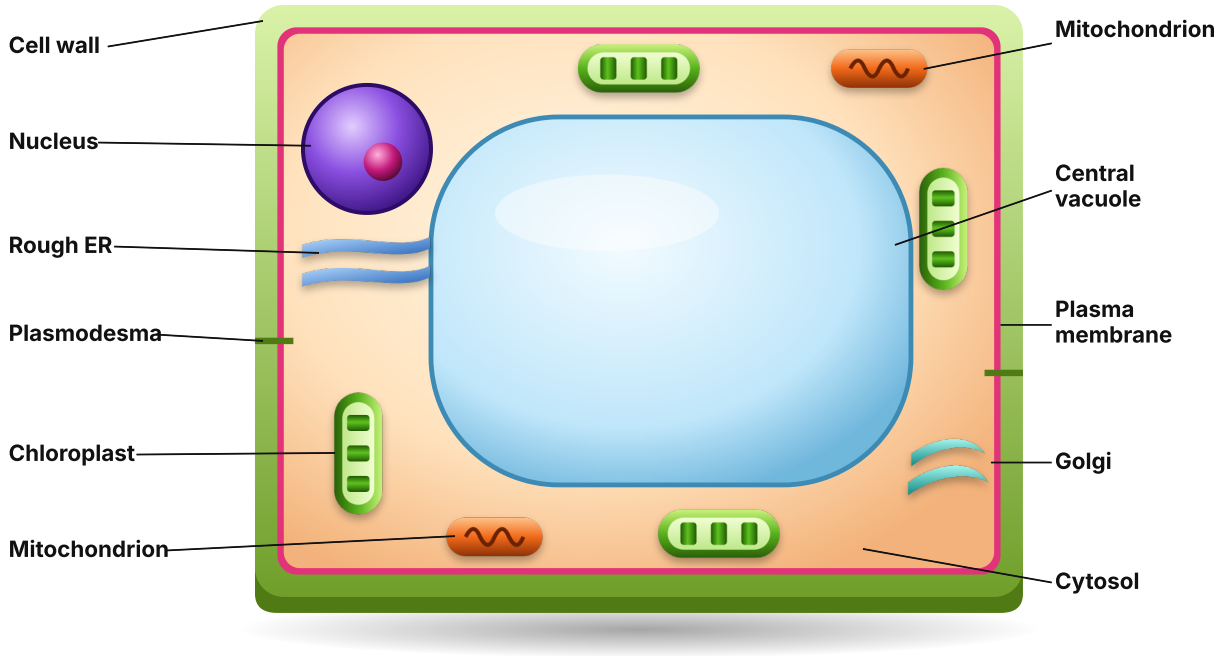


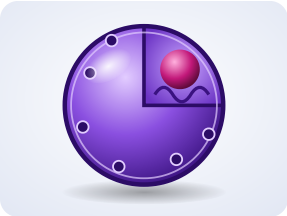
Figure 2.3 A 3D cutaway of a plant cell. Compared with an animal cell, it adds a cellulose **cell wall**, **chloroplasts**, and a large **central vacuole** that can fill most of the cell. Plasmodesmata are channels through the wall that connect neighboring cells. Plant cells also have mitochondria.

◆ **MISCONCEPTION ALERT**

Plant cells have mitochondria too. Chloroplasts capture light energy and make sugar, but plant cells still need mitochondria to break that sugar down and make most of their ATP through cellular respiration. On the exam, "plants have chloroplasts *instead of* mitochondria" is always wrong.

Step 4: The 3D Organelle Gallery

Each card shows one structure, the cells it is found in, its **structure**, and its **function**. Learn them in pairs: for every structure, say what it does and why its shape suits that job.



Nucleus

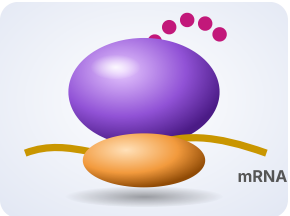
EUKARYOTES

STRUCTURE

Double-membrane **nuclear envelope** perforated by **nuclear pores**; contains chromatin (DNA + proteins) and the **nucleolus**.

FUNCTION

Stores and protects DNA; site of transcription. The nucleolus assembles ribosomal subunits. Pores let mRNA out and proteins in.



Ribosome

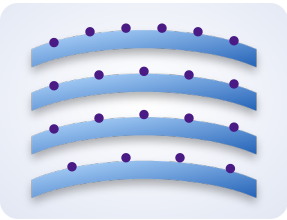
ALL CELLS (NO MEMBRANE)

STRUCTURE

Two subunits (large and small) made of **ribosomal RNA (rRNA)** and protein. Not surrounded by a membrane.

FUNCTION

Synthesizes proteins by translating mRNA. **Free** ribosomes make proteins used in the cytosol; **bound** ribosomes on rough ER make proteins for membranes, lysosomes, or secretion.



Rough Endoplasmic Reticulum

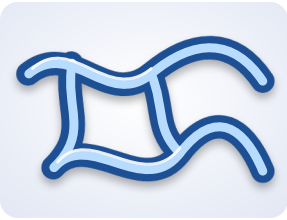
EUKARYOTES

STRUCTURE

Network of flattened membrane sacs (cisternae), continuous with the nuclear envelope, studded with ribosomes.

FUNCTION

Compartmentalizes the cell; ribosomes make proteins that enter the ER lumen to be folded, modified, and shipped in vesicles to the Golgi.



Smooth Endoplasmic Reticulum

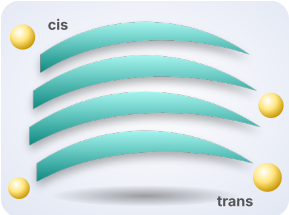
EUKARYOTES

STRUCTURE

Network of membrane tubules with **no ribosomes**.

FUNCTION

Synthesizes lipids (phospholipids, steroids), detoxifies drugs and poisons (especially in liver cells), and stores calcium ions (in muscle cells, it triggers contraction).



Golgi Complex

EUKARYOTES

STRUCTURE

Stack of flattened membrane sacs with a receiving (*cis*) face and a shipping (*trans*) face.

FUNCTION

Modifies, sorts, tags, and packages proteins and lipids from the ER (for example, adding sugars to make glycoproteins); sends them out in vesicles.



Mitochondrion

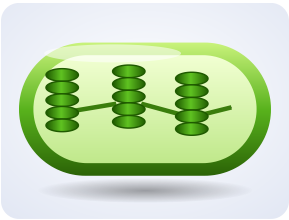
NEARLY ALL EUKARYOTES (PLANTS TOO)

STRUCTURE

Smooth **outer membrane** and highly folded **inner membrane** (folds = **cristae**) surrounding the fluid **matrix**; has its own circular DNA and ribosomes.

FUNCTION

Site of aerobic cellular respiration; makes most of the cell's ATP. Cristae increase the surface area for ATP production.



Chloroplast

PLANTS AND ALGAE

STRUCTURE

Double membrane surrounding fluid **stroma**; inside are stacked membrane sacs called **thylakoids** (a stack = **granum**); has its own DNA and ribosomes.

FUNCTION

Site of photosynthesis. Light reactions occur in thylakoid membranes; the Calvin cycle occurs in the stroma.



Lysosome

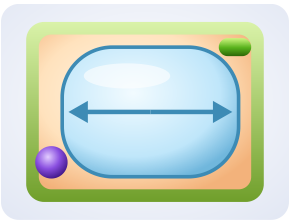
ANIMAL CELLS (MAINLY)

STRUCTURE

Membrane sac containing **hydrolytic enzymes** that work best at an acidic pH (about 5).

FUNCTION

Digests macromolecules, worn-out organelles (**autophagy**), and engulfed particles or bacteria; can trigger programmed cell death (**apoptosis**).



Central Vacuole

PLANTS (LARGE); SMALL VACUOLES IN OTHER CELLS

STRUCTURE

Large membrane-bound sac (membrane = **tonoplast**) filled with water, ions, pigments, and waste.

FUNCTION

Stores water and solutes; pushes the cytoplasm against the wall to create **turgor pressure** that keeps the plant firm; may hold toxins that deter herbivores.



Peroxisome

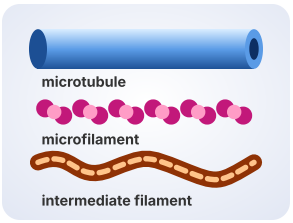
EUKARYOTES

STRUCTURE

Single-membrane sac containing oxidative enzymes, often with a crystalline core.

FUNCTION

Breaks down fatty acids and toxins; produces hydrogen peroxide (H_2O_2) and uses the enzyme **catalase** to convert it to water and oxygen.



Cytoskeleton

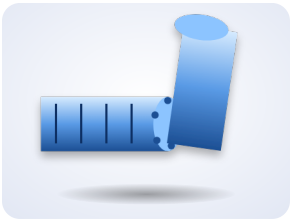
EUKARYOTES (SIMILAR PROTEINS IN PROKARYOTES)

STRUCTURE

Network of protein fibers: hollow **microtubules** (tubulin), thin **microfilaments** (actin), and rope-like **intermediate filaments**.

FUNCTION

Supports cell shape; anchors organelles; tracks for vesicle movement; moves chromosomes during division; forms cilia and flagella (microtubules); muscle contraction (actin).



Centrosome and Centrioles

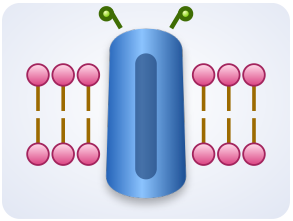
ANIMAL CELLS

STRUCTURE

A region near the nucleus containing a pair of **centrioles** at right angles, each built of nine triplets of microtubules.

FUNCTION

Organizes microtubules; helps form the spindle that separates chromosomes during cell division.



Plasma Membrane

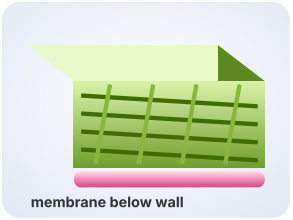
ALL CELLS

STRUCTURE

Phospholipid bilayer with embedded proteins, cholesterol (animals), and carbohydrate chains (fluid mosaic model).

FUNCTION

Selectively permeable boundary; controls transport; receives signals; cell recognition (Topics 2.4–2.9).



Cell Wall

PLANTS, FUNGI, MOST PROKARYOTES

STRUCTURE

Rigid layer *outside* the plasma membrane: cellulose fibers (plants), chitin (fungi), or peptidoglycan (bacteria).

FUNCTION

Provides shape and support; protects the cell; resists bursting when water rushes in. It is porous, so it does *not* control what enters.

Step 5: Plant vs. Animal Cells

STRUCTURE	ANIMAL CELL	PLANT CELL
Plasma membrane, nucleus, ribosomes, ER, Golgi, mitochondria, peroxisomes, cytoskeleton	Yes	Yes
Cell wall	No	Yes (cellulose)
Chloroplasts	No	Yes (in green tissues)
Central vacuole	No (small vacuoles only)	Yes, large
Lysosomes	Yes	Rare; the central vacuole performs digestion
Centrioles	Yes	No (most plants)
Cell junctions	Tight junctions, desmosomes, gap junctions	Plasmodesmata

EXAM TIP

Ribosomes are the universal organelle. Every living cell has ribosomes made of rRNA and protein, and the genetic code they read is nearly universal. When a question asks for evidence that all organisms share a **common ancestor**, ribosomes (along with DNA as genetic material and the plasma membrane) are a strong answer.

KEY TERMS

Prokaryotic cell
A cell without a nucleus or membrane-bound organelles (bacteria and archaea).

Eukaryotic cell
A cell with a nucleus and membrane-bound organelles.

Nucleoid
Region of a prokaryotic cell where its DNA is located.

Cytosol
The fluid portion of the cytoplasm.

Organelle
A specialized structure that performs a specific job within a cell.

Ribosome
rRNA–protein complex that synthesizes proteins.

Endomembrane system
The nuclear envelope, ER, Golgi, lysosomes, vacuoles, vesicles, and plasma membrane.

Cristae
Folds of the inner mitochondrial membrane.

Thylakoid
Membrane sac inside a chloroplast where light reactions occur.

Stroma
Fluid inside the chloroplast around the thylakoids.

Matrix
Fluid inside the inner mitochondrial membrane.

Hydrolytic enzyme
An enzyme that breaks bonds by hydrolysis.

Check Your Understanding 2.1

1. Which of the following structures is found in both prokaryotic and eukaryotic cells?

(A) Nucleus
(B) Mitochondria
(C) Ribosomes
(D) Golgi complex

2. A cell is observed to have a cell wall, a nucleus, mitochondria, and a large central vacuole, but no chloroplasts. This cell is most likely

(A) a bacterial cell
(B) an animal muscle cell
(C) a plant root cell
(D) a red blood cell

3. Which pair correctly matches an organelle with its function?

(A) Smooth ER — synthesis of lipids and detoxification
(B) Golgi complex — synthesis of ATP
(C) Lysosome — photosynthesis
(D) Nucleolus — digestion of worn-out organelles

4. A toxin inactivates the enzymes inside lysosomes. Which process would be most directly disrupted?

(A) Translation of mRNA

(B) Breakdown of engulfed bacteria and recycling of damaged organelles

(C) Synthesis of phospholipids

(D) Production of ATP by the electron transport chain

5. Which observation provides the strongest evidence that prokaryotes and eukaryotes share a common ancestor?

(A) Both types of cells have a nucleus.

(B) Both types of cells have ribosomes made of rRNA and protein.

(C) Both types of cells have chloroplasts.

(D) Both types of cells are the same size.

6. A plant cell contains both chloroplasts and mitochondria. Which statement best explains why?

(A) Chloroplasts make ATP for the cell at night, and mitochondria make it during the day.

(B) Mitochondria make sugar that chloroplasts then store.

(C) Mitochondria are needed only to support the cell wall.

(D) Chloroplasts convert light energy into chemical energy in sugars; mitochondria break down those sugars to produce ATP.

PART 1 · THE LESSONS

2.2

TOPIC 2.2

Cell Structure and Function

LEARNING TARGETS

- Explain how the structure of each organelle supports its function.
- Trace the path of a secreted protein through the endomembrane system, step by step.
- Explain how folded internal membranes increase surface area for reactions.
- Predict how damage to an organelle would affect the cell.

Structure Fits Function

Topic 2.1 asked *what* each organelle does. Topic 2.2 asks *how its structure makes that job possible*. The same idea from Unit 1 applies: **structure determines function**. A good answer always names a specific structural feature and then explains how that feature helps.

The Endomembrane System

The **endomembrane system** is a group of membranes and organelles that work together to make, modify, package, and transport proteins and lipids. It includes the **nuclear envelope, rough and smooth ER, Golgi complex, lysosomes, vacuoles, vesicles, and the plasma membrane**. These parts are connected either directly (the ER is continuous with the nuclear envelope) or by **vesicles**, small membrane sacs that bud off one membrane and fuse with another. Mitochondria and chloroplasts are *not* part of the endomembrane system.

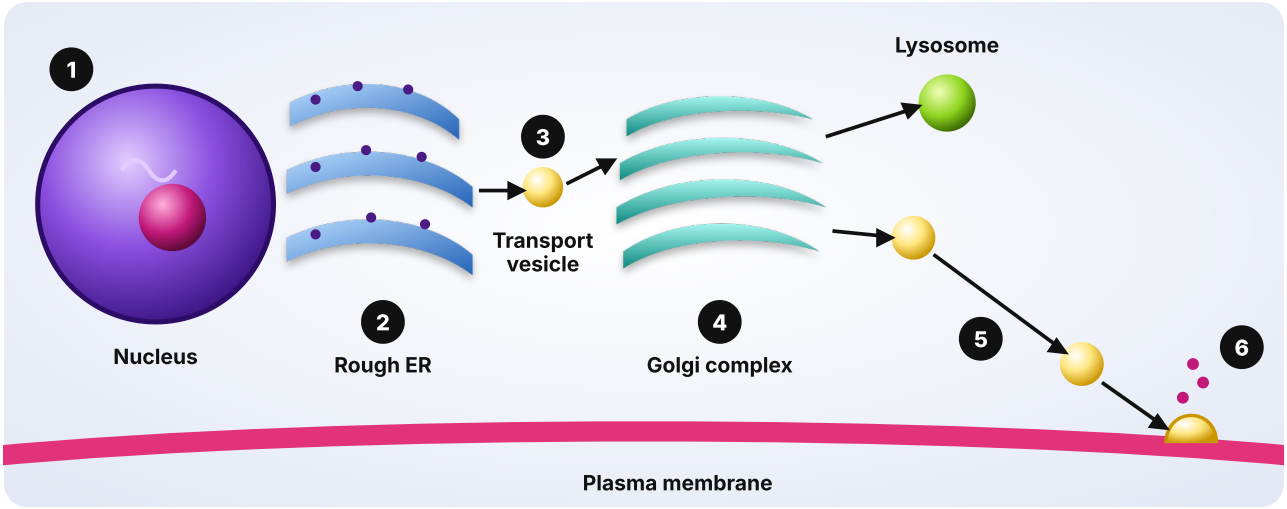


Figure 2.4 The secretory pathway through the endomembrane system. Numbers match the steps in the text. Some vesicles leaving the Golgi become lysosomes instead of being secreted.

Step by Step: How a Cell Makes and Secretes a Protein

Follow a digestive enzyme or a hormone such as insulin from gene to secretion:

1. **Nucleus.** The gene is transcribed into mRNA, which leaves through a nuclear pore.
2. **Rough ER.** A ribosome attaches to the ER and translates the mRNA. The growing polypeptide is threaded into the ER **lumen**, where it folds and may have sugars added.
3. **Transport vesicle.** The protein is packaged into a vesicle that buds off the ER and travels to the Golgi along cytoskeleton tracks.
4. **Golgi complex.** The vesicle fuses with the *cis* face. The protein is modified (for example, its sugar chains are trimmed or tagged), sorted, and shipped from the *trans* face.
5. **Secretory vesicle.** A vesicle carrying the finished protein moves to the plasma membrane.
6. **Exocytosis.** The vesicle fuses with the plasma membrane and releases the protein outside the cell. The vesicle membrane becomes part of the plasma membrane.

EXAM TIP

A cell that secretes large amounts of protein—pancreatic cells that make digestive enzymes or insulin, plasma cells that make antibodies—has an unusually large amount of **rough ER and Golgi**. A cell that makes steroid hormones (cells of the ovary or testis) or detoxifies drugs (liver cells) has a large amount of **smooth ER**. A cell with high energy demand (muscle, sperm) is packed with **mitochondria**. Use this logic to predict a cell's job from its organelles, or its organelles from its job.

Structure–Function Highlights

ORGANELLE	KEY STRUCTURAL FEATURE	HOW IT HELPS THE FUNCTION
Rough ER	Ribosomes on the surface; enclosed lumen	Proteins are made directly into a compartment where they can fold and be modified, separated from the cytosol.
Golgi complex	Separate cisternae with *cis* and *trans* faces	Allows step-by-step modification in order, like an assembly line, and sorting to the correct destination.
Mitochondrion	Double membrane; inner membrane folded into cristae	Folding greatly increases the surface area for electron transport chain proteins and ATP synthase, so more ATP can be made. The double membrane creates two compartments, allowing a proton gradient between them (Unit 3).
Chloroplast	Stacked thylakoid membranes in the stroma	Stacking increases the membrane surface area for light-absorbing pigments and electron transport; the thylakoid space and stroma are separate compartments for the light reactions and Calvin cycle.
Lysosome	Membrane that keeps an acidic interior (pH ≈ 5)	Hydrolytic enzymes work best at low pH and are kept away from the rest of the cell. If a lysosome leaks into the cytosol (pH ≈ 7.2), its enzymes are much less active, which helps protect the cell.

ORGANELLE	KEY STRUCTURAL FEATURE	HOW IT HELPS THE FUNCTION
Central vacuole	Large sac that takes up water	Water entering the vacuole pushes against the cell wall, creating turgor pressure that supports the plant. It also increases cell size cheaply without making more cytoplasm.
Nucleus	Double membrane with pores	Protects DNA while pores control traffic: mRNA and ribosome subunits exit; proteins such as enzymes for DNA replication enter.

Why Folded Membranes Matter

Many important reactions happen **on membranes**, because the enzymes and transport proteins that carry them out are embedded in the membrane. The more membrane surface area a cell has, the more of these proteins it can hold. Folding a membrane—like the cristae of mitochondria or the stacked thylakoids of chloroplasts—packs a large surface area into a small volume. You will see the same idea again in Topic 2.3 (microvilli) and Unit 3 (electron transport chains).

◆ **MISCONCEPTION ALERT**

Ribosomes are not part of the endomembrane system. Ribosomes have no membrane. They can be *attached* to the rough ER, but free and bound ribosomes are identical; where a ribosome works depends on a signal at the start of the protein it is making. Also note that **mitochondria and chloroplasts** are membrane-bound but are *not* part of the endomembrane system—they are not connected to it by vesicles, and they arose differently (Topic 2.11).

■ **KEY TERMS**

Endomembrane system
Membranes that make, modify, and transport proteins and lipids, linked directly or by vesicles.

Lumen
The internal space of the ER or another membrane-bound compartment.

Vesicle
Small membrane sac that carries substances between compartments.

Cisternae
Flattened membrane sacs of the ER or Golgi.

Secretion
Release of substances, such as proteins, from a cell by exocytosis.

Turgor pressure
Pressure of the cell contents against the cell wall.

Check Your Understanding 2.2

1. Which sequence correctly shows the path of a protein that is secreted from a cell?

(A) Golgi → rough ER → vesicle → plasma membrane

(B) Rough ER → vesicle → Golgi → vesicle → plasma membrane

(C) Smooth ER → lysosome → Golgi → plasma membrane

(D) Free ribosome → nucleus → Golgi → plasma membrane

2. Cells in the pancreas secrete large amounts of digestive enzymes. Compared with a typical cell, these cells would be expected to contain unusually large amounts of

(A) rough ER and Golgi complex

(B) smooth ER and peroxisomes

(C) chloroplasts and central vacuoles

(D) centrioles and lysosomes

3. The inner membrane of a mitochondrion is highly folded. Which is the best explanation of how this structure supports mitochondrial function?

(A) Folding makes the mitochondrion smaller so it can move more easily.

(B) Folding prevents any molecules from entering the matrix.

(C) Folding allows the mitochondrion to store DNA.

(D) Folding increases the surface area available for the proteins that produce ATP.

4. Lysosomal enzymes are most active at pH 5. The cytosol has a pH of about 7.2. Which is the most likely benefit of this difference?

(A) Enzymes that leak from a broken lysosome are less active in the cytosol, which limits damage to the cell.

(B) The cytosol is acidic enough for all lysosomal digestion to happen outside the lysosome.

(C) Lysosomal enzymes are converted into lipids at pH 7.2.

(D) The difference allows lysosomes to produce ATP.

5. A drug prevents vesicles from fusing with the *cis* face of the Golgi complex. Which outcome is most likely?

(A) Proteins will accumulate in the cytosol after being made by free ribosomes.

(B) Proteins made on the rough ER will fail to reach the Golgi and will not be secreted.

(C) Transcription in the nucleus will stop immediately.

(D) Mitochondria will be unable to divide.

6. Liver cells that break down toxins and drugs have large amounts of which organelle?

(A) Smooth ER

(B) Chloroplasts

(C) Nucleolus

(D) Central vacuole

PART 1 · THE LESSONS

2.3

TOPIC 2.3

Cell Size

LEARNING TARGETS

- Calculate the surface area, volume, and surface area-to-volume ratio (SA:V) of cubes, spheres, and cylinders.
- Explain why a high SA:V ratio makes exchange with the environment more efficient.
- Explain how cells and organisms increase surface area (microvilli, root hairs, folding, flattening).
- Explain how SA:V affects heat exchange in organisms of different sizes.

Why Are Cells So Small?

Every cell must exchange materials with its environment: nutrients and oxygen must come in, and wastes and heat must go out. All of this exchange happens across the **plasma membrane**—the cell's **surface area**. But the amount of material a cell needs, and the waste it produces, depends on how much cytoplasm it has—its **volume**.

Here is the problem: as a cell grows, its volume increases *faster* than its surface area. Volume grows with the **cube** of the length (L^3), but surface area grows only with the **square** (L^2). A large cell would have too little membrane to supply all of its cytoplasm, and substances would have to diffuse a long way to reach the center. So cells stay small, and a **high SA:V ratio** means more efficient exchange.

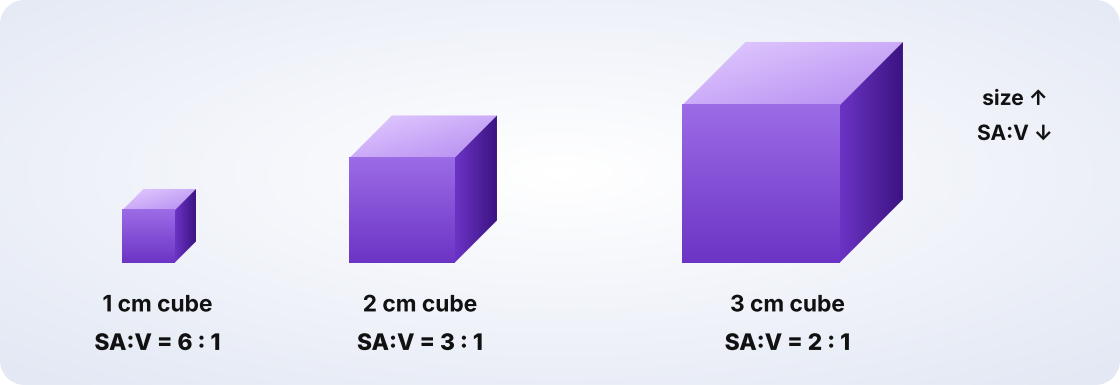


Figure 2.5 As a cube gets larger, its surface area-to-volume ratio gets smaller. A smaller "cell" has more membrane for every unit of volume it must supply.

The Formulas

The AP Biology Equations and Formulas sheet gives these formulas. You must be able to use them, but you do not need to memorize them.

SHAPE	SURFACE AREA (SA)	VOLUME (V)	VARIABLES
Cube	$SA = 6s^2$	$V = s^3$	s = length of one side
Sphere	$SA = 4\pi r^2$	$V = {}^4/_3\pi r^3$	r = radius
Cylinder	$SA = 2\pi rh + 2\pi r^2$	$V = \pi r^2 h$	r = radius, h = height
Rectangular solid	$SA = 2lh + 2lw + 2wh$	$V = lwh$	l, w, h = length, width, height

WORKED EXAMPLE 1 · COMPARING TWO CUBES

Problem: Cell A is a cube 2 µm on each side. Cell B is a cube 4 µm on each side. Which cell exchanges materials more efficiently?

1. **Surface area.** A: $6 \times 2^2 = 6 \times 4 =$ **24 µm²**. B: $6 \times 4^2 = 6 \times 16 =$ **96 µm²**.
2. **Volume.** A: $2^3 =$ **8 µm³**. B: $4^3 =$ **64 µm³**.
3. **Ratio (divide SA by V).** A: 24 ÷ 8 = **3** (3 : 1). B: 96 ÷ 64 = **1.5** (1.5 : 1).
4. **Interpret.** Cell A has twice as much membrane per unit volume, so it exchanges materials more efficiently. Doubling the side length halved the SA:V ratio.

WORKED EXAMPLE 2 · A SPHERICAL CELL

Problem: A spherical cell has a radius of 3 µm. Calculate its SA:V ratio. (Use $\pi \approx 3.14$.)

1. **Surface area.** $4 \times 3.14 \times 3^2 = 4 \times 3.14 \times 9 =$ **113.04 µm²**.
2. **Volume.** $(4/3) \times 3.14 \times 3^3 = (4/3) \times 3.14 \times 27 =$ **113.04 µm³**.
3. **Ratio.** 113.04 ÷ 113.04 = **1** (1 : 1).
4. **Shortcut.** For any sphere, SA:V = $3/r$. Here 3 ÷ 3 = 1. For a cube, SA:V = $6/s$. These shortcuts show directly that the ratio falls as size increases.

How Cells and Organisms Increase Surface Area

When a cell or organ needs a lot of exchange, it can increase surface area *without* greatly increasing volume:

- **Microvilli** — finger-like projections of the plasma membrane on intestinal cells greatly increase the surface area for absorbing nutrients.
- **Root hairs** — long, thin extensions of root epidermal cells increase the area for absorbing water and minerals.
- **Folded internal membranes** — cristae and thylakoids (Topic 2.2).
- **Flattened or elongated shapes** — a flat or long, thin cell has a higher SA:V ratio than a sphere of the same volume. Red blood cells are flattened disks; neurons are long and thin.
- **Organ-level structures** — the millions of alveoli in the lungs and the branched filaments of fish gills.

SA:V and Heat Exchange

The same math applies to whole organisms. Heat is produced by the volume of the body and lost through its surface. A **small animal**, such as a shrew, has a high SA:V ratio, so it loses heat rapidly and must have a **high metabolic rate per gram** to stay warm. A **large animal**, such as an elephant, has a low SA:V ratio, retains heat well, and has a lower metabolic rate per gram. Large animals in hot climates often have structures that add surface area for cooling, such as an elephant's large, thin ears.

◆ **MISCONCEPTION ALERT**

A larger cell has a *larger* surface area but a *smaller* SA:V ratio. Students often say "bigger cells have less surface area." That is false. The problem is that surface area does not keep up with volume. Always talk about the *ratio*.

● **EXAM TIP**

When asked to explain a size limit, give the full chain: **as size increases → volume increases faster than surface area → SA:V decreases → the membrane cannot exchange materials (or heat) fast enough to meet the needs of the volume.** Answers that stop at "the cell is too big" do not earn the point.

■ **KEY TERMS**

Surface area
The total area of the outside of an object; for a cell, the area of the plasma membrane.

Volume
The amount of space an object occupies; for a cell, the amount of cytoplasm.

SA:V ratio
Surface area divided by volume; higher values mean more efficient exchange.

Microvilli
Membrane projections that increase the surface area of absorptive cells.

Check Your Understanding 2.3

1. A cube-shaped cell has sides of 5 µm. What is its surface area-to-volume ratio?

(A) 0.6 : 1
(B) 1.2 : 1
(C) 5 : 1
(D) 6 : 1

2. As a cell grows larger without changing shape, which statement is true?

(A) Its surface area decreases while its volume increases.
(B) Its surface area and volume increase at the same rate.
(C) Its volume increases faster than its surface area, so its SA:V ratio decreases.
(D) Its SA:V ratio increases, making exchange more efficient.

3. Cells that line the small intestine have thousands of microvilli. The main advantage of this structure is that it

(A) increases the volume of the cell so it can store more nutrients

(B) increases the surface area of the membrane available for absorption

(C) decreases the SA:V ratio so the cell loses less water

(D) provides the cell with a cell wall

4. Which of the following organisms would be expected to have the highest metabolic rate per gram of body mass?

(A) An elephant

(B) A horse

(C) A human

(D) A mouse

5. Three cells have the same volume. Which shape gives the highest SA:V ratio?

(A) A sphere

(B) A cube

(C) A long, thin cylinder

(D) All three have the same SA:V ratio.

6. Agar cubes containing a pH indicator are placed in vinegar, which diffuses inward and changes the color of the agar. After 5 minutes, which cube will have the greatest *percentage* of its volume changed in color?

(A) A 1 cm cube

(B) A 2 cm cube

(C) A 3 cm cube

(D) All cubes will have the same percentage changed.

PART 1 · THE LESSONS

2.4

TOPIC 2.4

Plasma Membranes

LEARNING TARGETS

- Describe the structure of the plasma membrane using the fluid mosaic model.
- Explain how the amphipathic nature of phospholipids causes them to form a bilayer.
- Describe the roles of integral and peripheral proteins, cholesterol, glycoproteins, and glycolipids.
- Explain the factors that affect membrane fluidity.

Step 1: Phospholipids Form a Bilayer

Recall from Unit 1 that a phospholipid is **amphipathic**: it has a **hydrophilic** (polar) phosphate head and two **hydrophobic** (nonpolar) fatty acid tails. In water, phospholipids arrange themselves so that the heads face the watery environment on both sides (the cytosol inside and the extracellular fluid outside) and the tails face each other in the middle, away from water. The result is a **phospholipid bilayer** with a hydrophobic core. No energy is needed to build it; hydrophobic interactions drive it to form on its own.

Step 2: The Fluid Mosaic Model

The plasma membrane is described by the **fluid mosaic model**:

- **Fluid** — the phospholipids and many proteins are not locked in place. They drift sideways within their layer, so the membrane behaves like a thin, flexible fluid rather than a rigid wall.
- **Mosaic** — the membrane is a patchwork of many different molecules: phospholipids, several kinds of proteins, cholesterol, and carbohydrates.

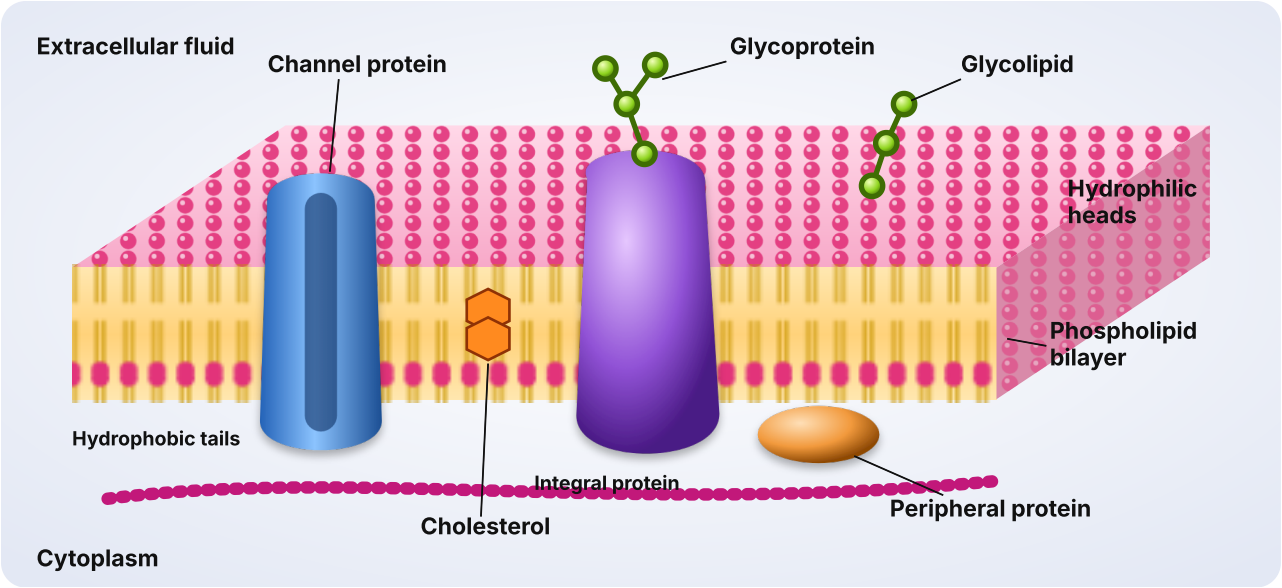


Figure 2.6 The fluid mosaic model. Integral proteins span the bilayer; peripheral proteins sit on its surface. Carbohydrate chains on glycoproteins and glycolipids face the outside of the cell. Cholesterol is wedged between the phospholipid tails in animal cells.

Step 3: Membrane Proteins

Proteins carry out most of the membrane's specific functions. They are classified by how they sit in the membrane:

- **Integral proteins** are embedded in the bilayer. Many are **transmembrane proteins** that span the whole membrane. The parts inside the hydrophobic core have mostly **nonpolar** amino acids; the parts that stick out into water have mostly **polar or charged** amino acids. This is Unit 1 chemistry at work.
- **Peripheral proteins** are loosely attached to the surface of the membrane or to integral proteins. They are not embedded in the hydrophobic core.

Membrane proteins perform six main jobs: **transport** (channels, carriers, pumps), **enzymatic activity**, **signal reception** (receptors), **cell–cell recognition**, **intercellular joining** (junctions), and **attachment** to the cytoskeleton and extracellular matrix.

Step 4: Carbohydrates and Cholesterol

Glycoproteins (proteins with attached carbohydrate chains) and **glycolipids** (lipids with attached carbohydrates) are found on the *outer* surface of the plasma membrane. Their sugar chains act as identification tags for **cell–cell recognition**: the immune system uses them to tell "self" from "nonself," and the ABO blood types are determined by them.

Cholesterol, a steroid found in animal cell membranes, acts as a **fluidity buffer**. At warm temperatures it restrains the movement of phospholipids, making the membrane less fluid; at cold temperatures it prevents the tails from packing tightly, keeping the membrane from becoming too rigid.

Step 5: What Controls Fluidity?

FACTOR	EFFECT ON FLUIDITY	WHY
Higher temperature	More fluid	Molecules move faster.
More unsaturated fatty acid tails	More fluid	C=C double bonds put kinks in the tails, so they cannot pack tightly.
More saturated fatty acid tails	Less fluid	Straight tails pack tightly together.
Cholesterol	Buffers fluidity	Reduces fluidity when warm; prevents solidifying when cold.

Fluidity matters because membrane proteins must be able to move and change shape to work. Organisms adjust their membranes to their environment: plants and bacteria in cold climates increase the proportion of unsaturated fatty acids in their membranes.

Beyond the Membrane: Cell Walls and the Extracellular Matrix

Plants, fungi, and most prokaryotes have a **cell wall** outside the plasma membrane: cellulose in plants, chitin in fungi, and peptidoglycan in bacteria. Animal cells lack walls but are surrounded by an **extracellular matrix (ECM)** of glycoproteins such as collagen, which supports the cell and helps it communicate with its neighbors.

◆ **MISCONCEPTION ALERT**

The membrane is not a static, solid layer. "Fluid" means the components move laterally. Also, phospholipids rarely flip from one layer to the other, so the two layers can have different compositions—for example, carbohydrate chains are only on the outer face.

■ **KEY TERMS**

Fluid mosaic model
Model of the membrane as a fluid bilayer with a mosaic of embedded and attached molecules.

Amphipathic
Having both hydrophilic and hydrophobic regions.

Integral protein
A protein embedded in the bilayer; transmembrane proteins span it.

Peripheral protein
A protein loosely bound to the membrane surface.

Glycoprotein
A protein with attached carbohydrate chains.

Glycolipid
A lipid with attached carbohydrate chains.

Cholesterol
Steroid in animal membranes that buffers fluidity.

Extracellular matrix
Network of glycoproteins outside animal cells.

Check Your Understanding 2.4

1. Phospholipids spontaneously form a bilayer in water because

(A) their hydrophobic tails are attracted to water

(B) their hydrophilic heads face water while their hydrophobic tails are shielded from it

(C) they form covalent bonds with one another

(D) ATP is used to assemble them

2. A transmembrane protein crosses the bilayer once. The region of the protein inside the bilayer most likely consists of amino acids with

(A) charged R groups

(B) polar R groups

(C) nonpolar R groups

(D) no R groups

3. Which membrane components are most directly involved in cell–cell recognition?

(A) Glycoproteins and glycolipids

(B) Cholesterol molecules

(C) Fatty acid tails

(D) Peripheral proteins on the cytoplasmic side

4. A species of fish lives in very cold water. Compared with a related species in warm water, its membranes would most likely contain

(A) more saturated fatty acids

(B) more unsaturated fatty acids

(C) no phospholipids

(D) no membrane proteins

5. Which statement best describes the role of cholesterol in an animal cell membrane?

(A) It makes the membrane permeable to ions.

(B) It forms channels for water.

(C) It binds to signal molecules outside the cell.

(D) It reduces fluidity at high temperatures and prevents tight packing at low temperatures.

PART 1 · THE LESSONS

2.5

TOPIC 2.5

Membrane Permeability

LEARNING TARGETS

- Explain what it means for a membrane to be selectively permeable.
- Predict whether a substance can diffuse directly across the bilayer based on its size, polarity, and charge.
- Explain why polar molecules and ions need transport proteins.
- Describe the role of the cell wall in permeability and support.

Selective Permeability

The plasma membrane is **selectively permeable**: some substances cross it easily, while others cross slowly or not at all without help. This property lets the cell keep its internal environment different from the outside—an essential feature of life.

Whether a substance can slip directly through the bilayer depends on the **hydrophobic core**. Use this step-by-step decision:

1. **Is it small and nonpolar?** (O_2, CO_2, N_2, steroid hormones, other lipids) → It dissolves in the hydrophobic core and **diffuses freely**.
2. **Is it small and polar?** (water, ethanol, urea, glycerol) → It crosses **slowly**. Water crosses much faster through channel proteins called **aquaporins**.
3. **Is it large and polar?** (glucose, amino acids) → It **needs a transport protein**.
4. **Is it an ion?** (Na^+, K^+, Cl^-, H^+, Ca^{2+}) → Its charge (and its shell of water molecules) keeps it out of the hydrophobic core. It **needs a channel or carrier protein**.
5. **Is it a macromolecule?** (proteins, polysaccharides) → Too big for any protein; it crosses by **endocytosis or exocytosis** (Topic 2.9).

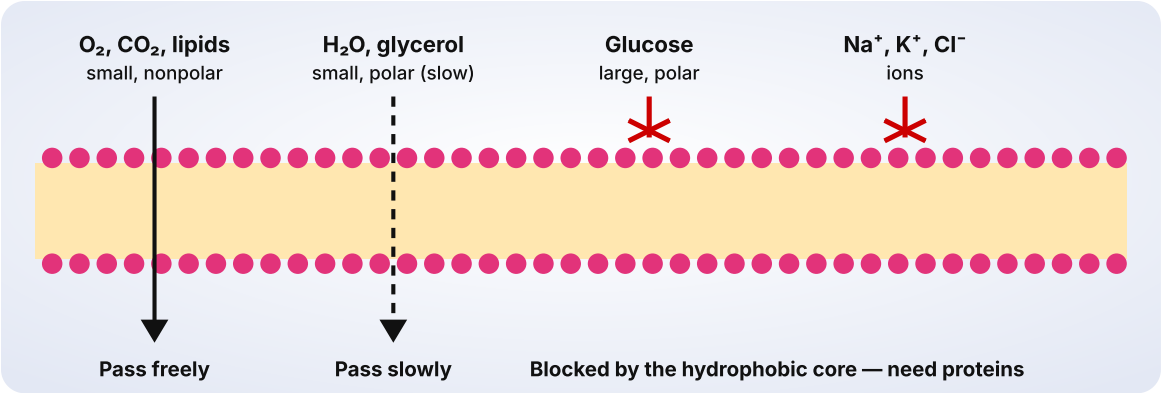


Figure 2.7 Permeability of a pure phospholipid bilayer. The hydrophobic core lets small nonpolar molecules through but blocks large polar molecules and ions.

The Role of the Cell Wall

A cell wall is **not** selectively permeable in the way a membrane is. It is porous and lets water and small solutes pass freely. Its jobs are **structural**: it maintains the cell's shape, protects it, and keeps it from bursting when water enters. The plasma membrane, just inside the wall, is still the structure that controls what enters and leaves the cell.

EXAM TIP

When asked to predict whether a molecule will cross the membrane, name the property and the reason: "O_2 diffuses directly across the membrane *because* it is small and nonpolar, so it can pass through the hydrophobic interior of the phospholipid bilayer." For ions: "Na^+ requires a channel or carrier protein *because* its charge prevents it from passing through the hydrophobic core."

KEY TERMS

Selective permeability
Property of a membrane that allows some substances to cross more easily than others.

Aquaporin
Channel protein that allows rapid movement of water.

Transport protein
A membrane protein that helps a substance cross.

Cell wall
Rigid, porous structural layer outside the plasma membrane.

Check Your Understanding 2.5

1. Which molecule would cross a pure phospholipid bilayer most rapidly?

(A) K^+

(B) Glucose

(C) O_2

(D) An amino acid

2. Steroid hormones such as testosterone can enter cells without the help of a transport protein. The best explanation is that steroids are

(A) charged and attracted to phosphate heads

(B) nonpolar and can pass through the hydrophobic interior of the membrane

(C) larger than glucose

(D) made of amino acids

3. Why do ions such as Na^+ require transport proteins to cross the plasma membrane?

(A) Their charge prevents them from passing through the hydrophobic core of the bilayer.

(B) They are too large to pass between phospholipids.

(C) They are nonpolar.

(D) They are made by the Golgi complex.

4. Which statement about the plant cell wall is correct?

(A) It is selectively permeable and controls which ions enter the cell.

(B) It is made of phospholipids.

(C) It is porous and mainly provides structural support and protection against bursting.

(D) It replaces the plasma membrane in plant cells.

5. A researcher makes artificial vesicles from pure phospholipids and places them in a solution containing water and glucose. Compared with real cells, the vesicles would most likely

(A) take up glucose much more slowly, because they lack transport proteins

(B) take up glucose faster, because they have no proteins blocking the way

(C) be completely impermeable to water

(D) take up glucose at the same rate as real cells

PART 1 · THE LESSONS

2.6

TOPIC 2.6

Membrane Transport

LEARNING TARGETS

- Describe a concentration gradient and explain why it stores energy.
- Compare passive and active transport in terms of direction and energy.
- Explain how selective permeability lets cells maintain concentration gradients.
- Describe endocytosis and exocytosis as ways to move large substances.

Step 1: Concentration Gradients

A **concentration gradient** exists when a substance is more concentrated in one region than another. Molecules are always in random motion, so over time they spread out from where they are **more concentrated** to where they are **less concentrated**. This net movement is **diffusion**, and it continues until the concentration is equal everywhere (**dynamic equilibrium**—molecules still move, but there is no *net* change).

Moving "down" a gradient (high → low) happens on its own. Moving "up" a gradient (low → high) requires energy. Because a gradient is an uneven arrangement that would naturally even out, a gradient is a form of **stored potential energy**. Cells spend energy to build gradients and then use them to power work.

Step 2: Passive vs. Active Transport

	PASSIVE TRANSPORT	ACTIVE TRANSPORT
Direction	Down the concentration gradient (high → low)	Against the concentration gradient (low → high)
Energy	No energy input from the cell	Requires energy, usually ATP (or another gradient)
Proteins	None (simple diffusion) or channels and carriers (facilitated diffusion)	Always requires a carrier protein (pump)
Examples	O_2 diffusing into a cell; water moving through aquaporins (osmosis); glucose through GLUT carriers	Sodium–potassium pump; proton pumps; cotransport of glucose into intestinal cells
Result	Gradient becomes smaller	Gradient is created or maintained

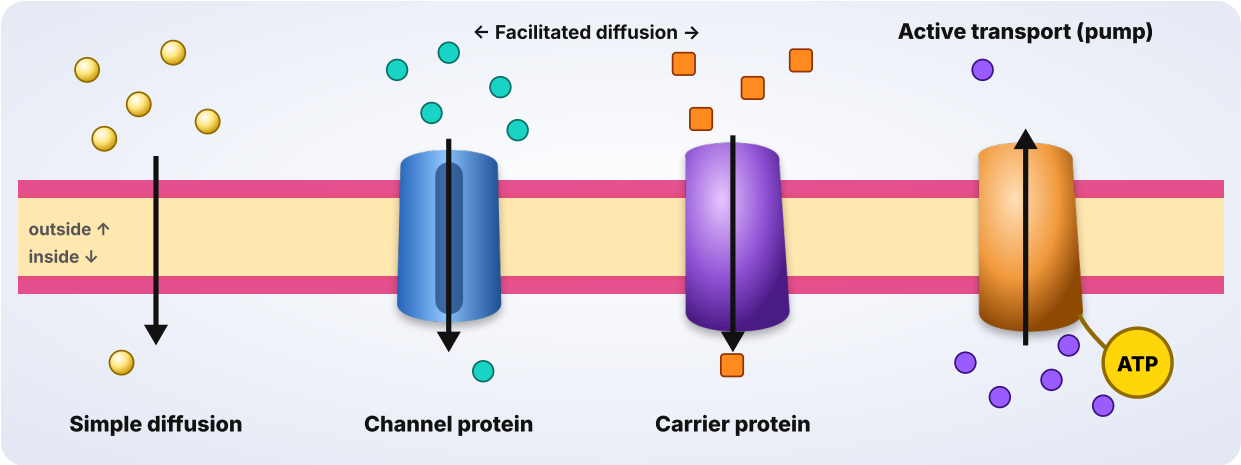


Figure 2.8 Four ways across the membrane. The three passive methods move substances *down* their gradients (high to low) without energy input. Active transport uses energy, often from ATP, to move a substance *up* its gradient (low to high).

Step 3: Moving Large Substances

Some molecules, such as proteins and polysaccharides, are far too large for any channel or carrier. Cells move them in bulk using vesicles. Both processes require energy.

- **Exocytosis** — a vesicle inside the cell fuses with the plasma membrane and releases its contents outside. Example: secretion of insulin or neurotransmitters.
- **Endocytosis** — the plasma membrane folds inward around material and pinches off to form a vesicle inside the cell. Example: a white blood cell engulfing a bacterium. Topic 2.9 describes the three kinds of endocytosis.

◆ **MISCONCEPTION ALERT**

"Passive" does not mean "without proteins." Facilitated diffusion uses proteins and is still passive, because the substance moves down its gradient and the cell spends no energy. The test for passive vs. active is *direction and energy*, not whether a protein is involved.

■ **KEY TERMS**

Concentration gradient
A difference in concentration of a substance between two regions.

Diffusion
Net movement of a substance from higher to lower concentration.

Dynamic equilibrium
State with no net change although molecules continue to move.

Passive transport
Movement down a gradient with no energy input from the cell.

Active transport
Movement against a gradient using energy, through a carrier protein.

Endocytosis
Uptake of material by folding in the plasma membrane to form a vesicle.

Exocytosis
Release of material when a vesicle fuses with the plasma membrane.

Check Your Understanding 2.6

1. Which feature distinguishes active transport from all forms of passive transport?

(A) Active transport uses membrane proteins.

(B) Active transport only moves water.

(C) Active transport moves substances against their concentration gradient using energy.

(D) Active transport only occurs in plant cells.

2. A cell contains 5 mM K^+ outside and 140 mM K^+ inside. To move additional K^+ into the cell, the cell must use

(A) simple diffusion

(B) osmosis

(C) active transport

(D) facilitated diffusion through a K^+ channel

3. A substance diffuses into a cell until its concentration is equal inside and outside. Which statement is true at this point?

(A) All movement of the substance stops.

(B) Molecules continue to cross in both directions, but there is no net movement.

(C) The substance begins to move only outward.

(D) The cell must spend ATP to keep the concentrations equal.

4. Nerve cells release neurotransmitter molecules stored in vesicles. This release occurs by

(A) exocytosis

(B) endocytosis

(C) osmosis

(D) simple diffusion

5. Why is a concentration gradient across a membrane considered a form of stored energy?

(A) Molecules in a gradient contain more covalent bonds.

(B) Gradients produce light energy.

(C) Gradients only form in the presence of ATP.

(D) The gradient can do work as molecules move down it, because the uneven distribution tends to even out.

PART 1 · THE LESSONS

2.7

TOPIC 2.7

Facilitated Diffusion

LEARNING TARGETS

- Explain how channel proteins and carrier proteins move polar molecules and ions down their gradients.
- Describe aquaporins and gated ion channels.
- Explain why the rate of facilitated diffusion levels off (saturates) at high concentrations.

Two Kinds of Helper Proteins

Facilitated diffusion is the passive movement of a substance down its concentration gradient *with the help of a transport protein*. It moves polar molecules and ions that cannot easily cross the hydrophobic core. Each transport protein is **specific**: its shape and the chemistry of its interior allow only certain substances to pass.

- **Channel proteins** form a hydrophilic tunnel through the membrane. Substances flow through rapidly. Examples:
  - **Aquaporins** — channels that let water molecules pass in single file, thousands of times faster than through the bilayer. Kidney cells and plant root cells have many.
  - **Ion channels** — let specific ions pass. Many are **gated**: they open or close in response to a signal, such as a change in voltage or the binding of a molecule. Gated channels are essential for nerve impulses and muscle contraction (Unit 4).
- **Carrier proteins** bind a specific substance, change shape, and release it on the other side. They are slower than channels. Example: **GLUT** glucose transporters move glucose into cells down its gradient.

Rate of Facilitated Diffusion

The rate of **simple diffusion** keeps increasing as the concentration gradient gets steeper. The rate of **facilitated diffusion** also increases at first, but then **levels off**. Why? There are a limited number of transport proteins. Once every carrier is busy (the carriers are **saturated**), adding more solute cannot speed things up. The only way to raise the maximum rate is to add more transport proteins to the membrane.

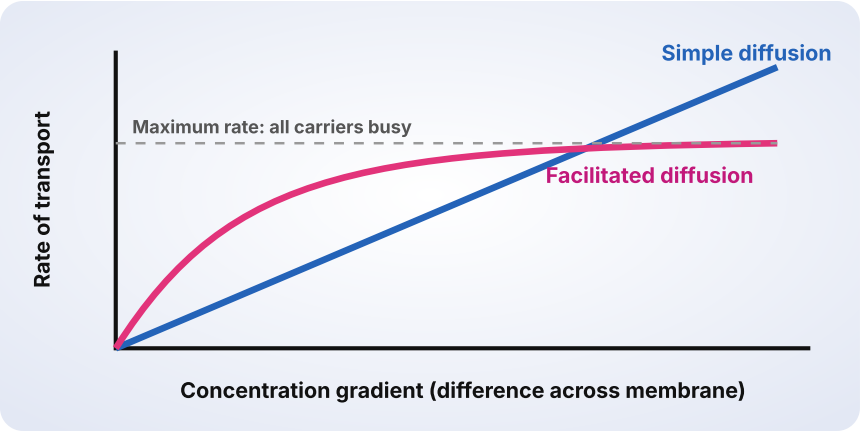


Figure 2.9 Facilitated diffusion saturates because the number of transport proteins is limited; simple diffusion does not.

EXAM TIP

A graph that **levels off** as concentration increases is the signature of a **protein-mediated** process (facilitated diffusion or active transport). If a question asks how to increase the maximum rate, the answer is **more transport proteins**, not a bigger gradient.

MISCONCEPTION ALERT

Facilitated diffusion can never move a substance against its gradient. No matter how many channels or carriers there are, the substance still only moves from high to low concentration. To go against the gradient, the cell needs active transport.

KEY TERMS

Facilitated diffusion
Passive movement of a substance down its gradient through a transport protein.

Channel protein
Protein with a hydrophilic pore that substances flow through.

Carrier protein
Protein that binds a substance and changes shape to move it across.

Gated channel
A channel that opens or closes in response to a stimulus.

Saturation
Point at which all transport proteins are in use and the rate cannot increase.

Specificity
A transport protein's ability to move only particular substances.

Check Your Understanding 2.7

1. Which statement correctly describes facilitated diffusion?

(A) It moves substances against their gradients using ATP.

(B) It moves substances down their gradients through specific transport proteins without energy input from the cell.

(C) It moves only nonpolar molecules.

(D) It requires vesicles.

2. Kidney cells that reabsorb large amounts of water would be expected to have many

(A) aquaporins

(B) sodium–potassium pumps only

(C) glycolipids

(D) lysosomes

3. The rate at which a sugar enters a cell increases as its outside concentration increases, but then levels off. The most likely explanation is that

(A) the sugar enters by simple diffusion

(B) the sugar is transported by a limited number of carrier proteins that become saturated

(C) the membrane becomes impermeable to water

(D) the sugar is being moved by exocytosis

4. A gated ion channel opens when a signal molecule binds to it. When the channel opens, ions will move

(A) in whichever direction ATP is used to move them

(B) against their concentration gradient

(C) down their concentration (electrochemical) gradient

(D) only out of the cell, regardless of concentration

5. Which change would increase the *maximum* rate of glucose uptake by facilitated diffusion?

(A) Adding more GLUT carrier proteins to the membrane

(B) Removing all cholesterol from the membrane

(C) Lowering the glucose concentration outside the cell

(D) Adding ATP to the cell

PART 1 · THE LESSONS

2.8

TOPIC 2.8

Tonicity and Osmoregulation

LEARNING TARGETS

- Define osmosis and predict the direction of water movement between two solutions.
- Use the terms hypertonic, hypotonic, and isotonic correctly, and predict what happens to animal and plant cells in each.
- Calculate solute potential and water potential and use them to predict water movement.
- Explain how organisms osmoregulate.

Step 1: Osmosis

Osmosis is the diffusion of **water** across a selectively permeable membrane. Water moves from the side with a **lower solute concentration** (more free water) to the side with a **higher solute concentration** (less free water). Put simply: **water follows solutes**. The membrane must be permeable to water but not to the solute; otherwise the solute itself would diffuse and even things out.

Step 2: Tonicity

Tonicity describes how a surrounding solution affects a cell's water content. The terms always *compare* two solutions, so always ask: "hypertonic *relative to what*?"

SOLUTION OUTSIDE IS…	MEANING	NET WATER MOVEMENT	ANIMAL CELL	PLANT CELL
Hypotonic	Lower solute concentration than the cell	Into the cell	Swells; may burst (**lysis**)	Swells against wall; **turgid** (ideal state)
Isotonic	Equal solute concentration	No net movement	Normal (ideal state)	Limp; **flaccid**
Hypertonic	Higher solute concentration than the cell	Out of the cell	Shrivels (**crenation**)	Membrane pulls away from wall; **plasmolysis**

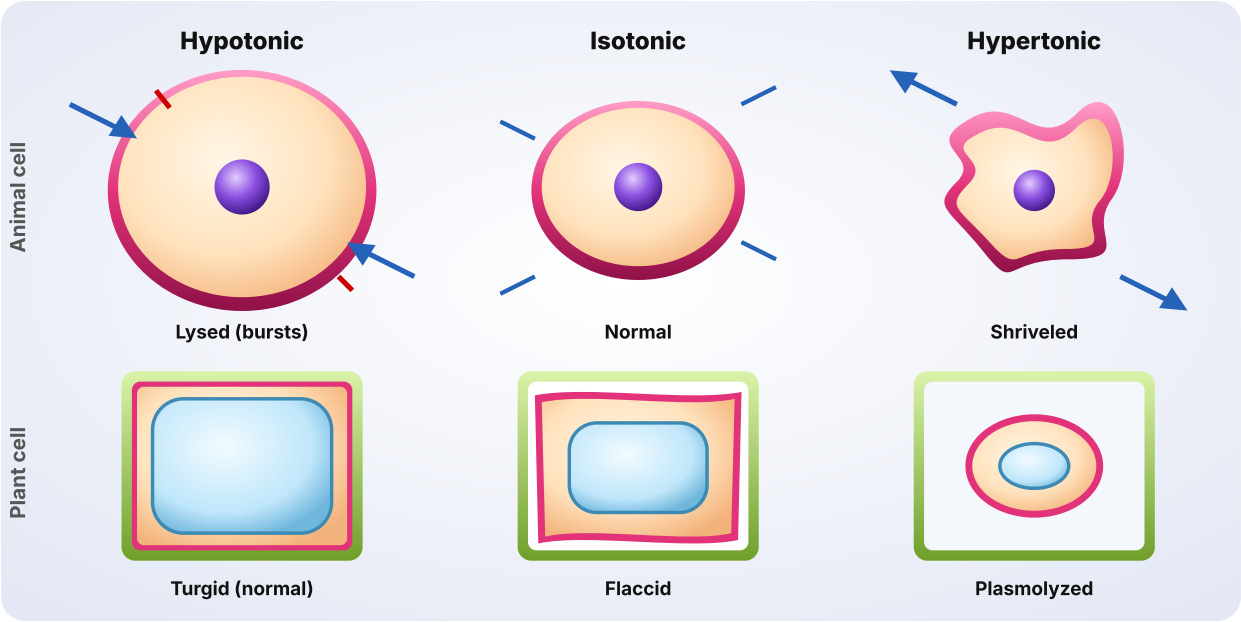


Figure 2.10 Animal and plant cells in solutions of different tonicity. Blue arrows show net water movement. The plant cell wall prevents bursting in a hypotonic solution, so the turgid state is ideal for plant cells, while the isotonic state is ideal for animal cells.

Step 3: Water Potential

Biologists predict the direction of water movement precisely using **water potential** (symbol Ψ, the Greek letter psi). Water potential measures the tendency of water to move. Its unit is the **bar** or the **megapascal** (1 MPa = 10 bars). The single rule to remember is:

> **Water always moves from higher water potential to lower water potential.**

Pure water at atmospheric pressure has Ψ = 0. Adding solute *lowers* water potential (makes it negative). Adding pressure *raises* it.

EQUATION	MEANING
$\Psi = \Psi_P + \Psi_S$	Water potential = pressure potential + solute potential.
$\Psi_S = -iCRT$	i = ionization constant (1.0 for sucrose and glucose, which do not ionize; 2.0 for NaCl, which splits into two ions); C = molar concentration (mol/L); R = pressure constant = 0.0831 L·bar/(mol·K); T = temperature in kelvins (°C + 273).
Ψ_P	Pressure potential. It is 0 in an open container; it is positive inside a turgid plant cell, where the wall pushes back on the contents.

WORKED EXAMPLE 1 · SOLUTE POTENTIAL OF A SUCROSE SOLUTION

Problem: Calculate the solute potential of a 0.20 M sucrose solution at 22°C in an open beaker.

1. **List the variables.** i = 1.0 (sucrose does not ionize); C = 0.20 mol/L; R = 0.0831; T = 22 + 273 = 295 K.
2. **Substitute.** $\Psi_S = -(1.0)(0.20)(0.0831)(295)$.
3. **Multiply.** 0.20 × 0.0831 = 0.01662; 0.01662 × 295 = 4.90. So Ψ_S = **−4.90 bars**.
4. **Find Ψ.** The beaker is open, so $\Psi_P = 0$ and $\Psi = 0 + (-4.90)$ = **−4.90 bars**.

WORKED EXAMPLE 2 · IONIC SOLUTE (NACL)

Problem: Calculate the solute potential of a 0.10 M NaCl solution at 25°C.

1. **Variables.** i = 2.0 ($NaCl \rightarrow Na^+ + Cl^-$); C = 0.10; R = 0.0831; T = 298 K.
2. **Calculate.** $\Psi_S = -(2.0)(0.10)(0.0831)(298)$ = **−4.95 bars**.
3. **Notice.** Because NaCl releases two particles per formula unit, a 0.10 M NaCl solution has almost the same solute potential as the 0.20 M sucrose solution above. Solute potential depends on the *number of dissolved particles*.

WORKED EXAMPLE 3 · WHICH WAY WILL WATER MOVE?

Problem: A plant cell has Ψ_S = −7.0 bars and Ψ_P = +3.0 bars. It is placed in a beaker of solution with Ψ = −2.0 bars. Which way does water move?

1. **Cell's water potential.** $\Psi = \Psi_P + \Psi_S = 3.0 + (-7.0)$ = **−4.0 bars**.
2. **Compare.** The solution is −2.0 bars; the cell is −4.0 bars. −2.0 is *higher* (less negative) than −4.0.
3. **Apply the rule.** Water moves from higher to lower Ψ, so water moves **into the cell**.
4. **Predict.** As water enters, the cell becomes more turgid and Ψ_P rises until the cell's Ψ equals −2.0 bars. Then there is no net movement.

Step 4: Reading an Osmosis Experiment

A classic lab places pieces of potato in sucrose solutions of different molarities and measures the percent change in mass. Percent change = (final mass − initial mass) ÷ initial mass × 100. A **gain** in mass means the solution was hypotonic to the potato; a **loss** means it was hypertonic. The concentration at which the graph crosses **0% change** is **isotonic** to the potato cells—it estimates the potato's solute concentration.

Step 5: Osmoregulation

Osmoregulation is the control of water balance and solute concentration in an organism. Examples:

- **Freshwater protists** such as *Paramecium* live in hypotonic water, so water constantly enters. A **contractile vacuole** collects excess water and pumps it out, using energy, to prevent bursting.
- **Plants** rely on hypotonic surroundings to stay turgid; a wilting plant has lost turgor pressure.

- **Freshwater fish** excrete large amounts of dilute urine and actively take up salt through their gills; **saltwater fish** drink seawater and actively excrete salt.
- In humans, the **kidneys** regulate the solute concentration of the blood, and IV fluids must be isotonic to blood cells.

◆ **MISCONCEPTION ALERT**

Salt does not "suck" water, and solutes do not move in osmosis—water does. Also, with negative numbers, "higher" means *closer to zero*: −2 bars is higher than −5 bars. Many students reverse this and lose the point.

● **EXAM TIP**

For any osmosis question, write the comparison explicitly: "The solution has a *higher solute concentration* (*lower water potential*) than the cell, so water moves out of the cell by osmosis." Naming both solute concentration and water potential shows full understanding.

■ **KEY TERMS**

Osmosis
Diffusion of water across a selectively permeable membrane.

Tonicity
The ability of a surrounding solution to cause a cell to gain or lose water.

Hypertonic
Having a higher solute concentration than another solution.

Hypotonic
Having a lower solute concentration than another solution.

Isotonic
Having an equal solute concentration.

Plasmolysis
Shrinking of a plant cell's cytoplasm away from its wall in a hypertonic solution.

Turgid / flaccid
Firm from water pressure against the wall / limp from lack of it.

Water potential (Ψ)
Measure of the tendency of water to move; water moves from high to low Ψ.

Solute potential (Ψ_S)
Component of Ψ due to dissolved solutes; always zero or negative.

Pressure potential (Ψ_P)
Component of Ψ due to physical pressure.

Osmoregulation
Control of water balance and solute concentration.

Contractile vacuole
Organelle that pumps excess water out of freshwater protists.

Check Your Understanding 2.8

1. A red blood cell is placed in distilled water. Which outcome is most likely?

(A) Water leaves the cell and it shrivels.

(B) Nothing happens because the cell is isotonic to water.

(C) Water enters the cell and it may burst.

(D) Solutes enter the cell by osmosis.

2. A plant cell placed in a concentrated salt solution undergoes

(A) plasmolysis

(B) lysis

(C) an increase in turgor pressure

(D) no change, because the cell wall blocks water

3. What is the solute potential of a 0.30 M glucose solution at 27°C? (R = 0.0831 L·bar/mol·K)

(A) −0.67 bars

(B) −7.48 bars

(C) −14.96 bars

(D) +7.48 bars

4. Cell X has a water potential of −0.8 MPa. It is placed in a solution with a water potential of −0.3 MPa. Which statement is correct?

(A) Water will move out of cell X because the solution has the higher water potential.

(B) There will be no net movement of water.

(C) Solutes will move into cell X by osmosis.

(D) Water will move into cell X because the solution has the higher water potential.

5. A freshwater *Paramecium* is treated with a drug that stops its contractile vacuole. Which outcome is most likely?

(A) The cell will shrink as water leaves.

(B) The cell will swell and may burst as water continues to enter.

(C) The cell will stay the same size because it is isotonic to fresh water.

(D) The cell will begin photosynthesis.

6. Potato cores were placed in sucrose solutions for 24 hours. The cores gained mass at 0.0 M and 0.2 M, lost mass at 0.4 M and 0.6 M, and showed the smallest change between 0.2 M and 0.4 M. The solute concentration of the potato cells is closest to

(A) 0.0 M

(B) 0.1 M

(C) 0.3 M

(D) 0.6 M

PART 1 · THE LESSONS

2.9

TOPIC 2.9

Mechanisms of Transport

LEARNING TARGETS

- Describe, step by step, how the sodium–potassium pump uses ATP to move ions against their gradients.
- Explain how the membrane potential and electrochemical gradients form.
- Explain how cotransport uses an existing gradient to move a second substance.
- Compare phagocytosis, pinocytosis, and receptor-mediated endocytosis.

Step 1: The Sodium–Potassium Pump

The **sodium–potassium pump** (Na^+/K^+ ATPase) is the best-known example of **primary active transport**—active transport powered directly by ATP. In animal cells it keeps Na^+ concentration **high outside** the cell and K^+ concentration **high inside**. Each cycle moves **3 Na^+ out** and **2 K^+ in**, both against their concentration gradients.

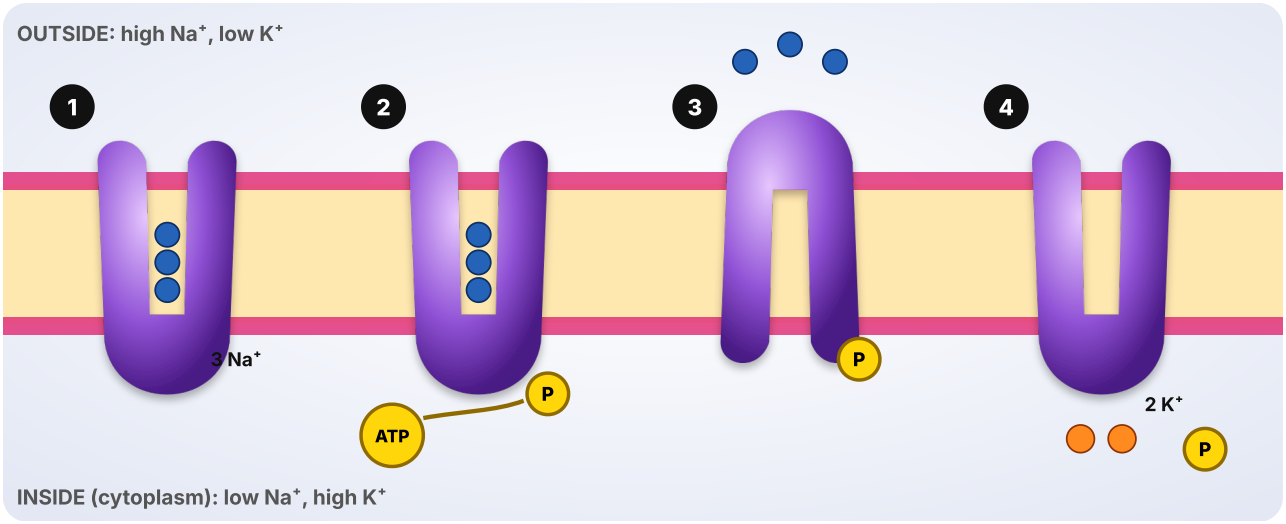


Figure 2.11 One cycle of the sodium–potassium pump. Blue circles are Na^+; orange circles are K^+. P is the phosphate group transferred from ATP. Numbers match the steps below.

1. **Na^+ binds.** The pump is open to the cytoplasm. Three Na^+ ions from inside the cell bind to it.
2. **ATP is used.** Binding of Na^+ triggers the pump to hydrolyze ATP. A phosphate group is attached to the pump (**phosphorylation**), and ADP is released.
3. **Shape change; Na^+ released outside.** Phosphorylation changes the pump's shape so it opens to the outside. Its affinity for Na^+ drops, and the three Na^+ ions are released outside the cell.

4 **K^+ binds; phosphate leaves; K^+ released inside.** Two K^+ ions from outside bind. This causes the phosphate to be released, the pump returns to its original shape, and the K^+ ions are released into the cytoplasm. The cycle repeats.

Step 2: Membrane Potential and Electrochemical Gradients

Because the pump moves three positive charges out for every two it brings in, and because K^+ leaks out through channels, the inside of a typical animal cell is **negative** relative to the outside—about −70 mV in a resting neuron. This voltage across the membrane is the **membrane potential**. An ion is therefore pushed by two forces: its **concentration gradient** and the **electrical gradient**. Together these make up the ion's **electrochemical gradient**. The membrane potential is the basis of nerve impulses (Unit 4).

Step 3: Cotransport (Secondary Active Transport)

A gradient built by a pump is stored energy. In **cotransport**, a carrier protein lets one substance flow *down* its gradient and uses that energy to drag a second substance *up* its gradient. No ATP is used by the cotransporter itself, but ATP was spent earlier by the pump that built the gradient, so this is called **secondary active transport**.

- **Intestinal cells:** the **sodium–glucose cotransporter** lets Na^+ flow into the cell (down its gradient, maintained by the Na^+/K^+ pump) and brings glucose in with it, even when glucose is more concentrated inside.
- **Plant cells:** a **proton pump** uses ATP to push H^+ out of the cell. The H^+ gradient then powers the **H^+/sucrose cotransporter**, which loads sucrose into cells for transport through the phloem.

Step 4: Bulk Transport with Vesicles

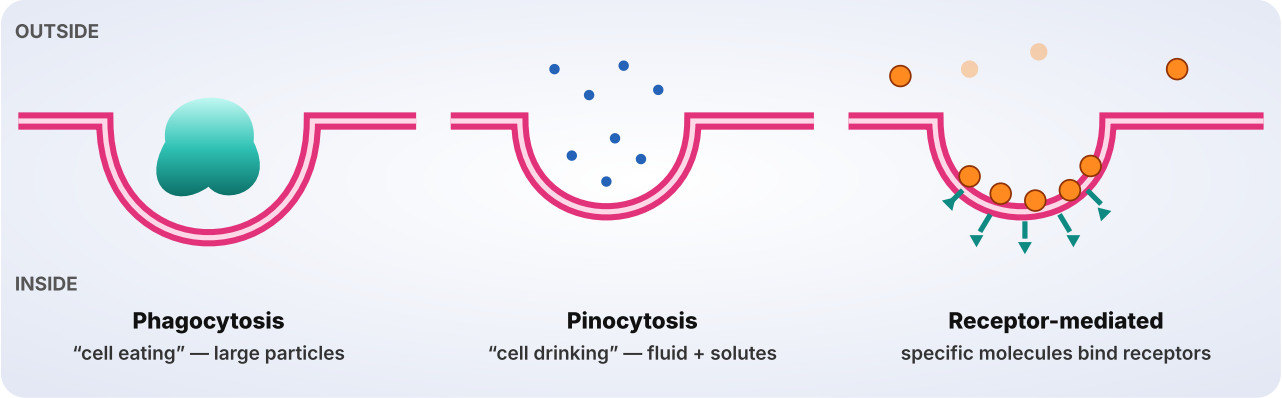


Figure 2.12 Three kinds of endocytosis. In each, the membrane folds inward and pinches off to form a vesicle. Receptor-mediated endocytosis is the most specific: only molecules that bind the receptors are taken in.

PROCESS	WHAT IS MOVED	EXAMPLE
Phagocytosis	Large particles or whole cells; the vesicle fuses with a lysosome for digestion	White blood cells engulfing bacteria; amoebas feeding
Pinocytosis	Droplets of extracellular fluid and whatever solutes they contain (nonspecific)	Many cells sampling their surroundings

PROCESS	WHAT IS MOVED	EXAMPLE
Receptor-mediated endocytosis	Specific molecules that bind receptor proteins clustered on the membrane	Uptake of cholesterol carried in LDL particles
Exocytosis	Contents of vesicles released outside the cell	Secretion of insulin, digestive enzymes, neurotransmitters

EXAM TIP

If a question says a transport process **stops when ATP production is blocked** (for example, with a poison such as cyanide), the process is active transport or bulk transport. If transport **continues** without ATP but still needs a protein, it is facilitated diffusion. Cotransport stops *eventually*, once the pump-maintained gradient runs down.

MISCONCEPTION ALERT

The sodium–potassium pump is not a channel. Ions never flow freely through it. It is a carrier protein that changes shape, and it always moves Na^+ *out* and K^+ *in*, against their gradients. Ions flowing *down* their gradients through open channels is facilitated diffusion.

KEY TERMS

Sodium–potassium pump
Carrier that uses ATP to move 3 Na^+ out and 2 K^+ in.

Primary active transport
Active transport powered directly by ATP.

Membrane potential
Voltage across a membrane due to unequal distribution of ions.

Electrochemical gradient
Combined concentration and electrical gradient for an ion.

Proton pump
Active transporter that pumps H^+ across a membrane.

Cotransport
Coupled movement of a substance up its gradient with another moving down its gradient.

Phagocytosis
Endocytosis of large particles (“cell eating”).

Pinocytosis
Nonspecific endocytosis of fluid (“cell drinking”).

Receptor-mediated endocytosis
Endocytosis of specific molecules bound to receptors.

Check Your Understanding 2.9

1. In each cycle, the sodium–potassium pump moves

(A) 3 Na^+ into the cell and 2 K^+ out of the cell

(B) 2 Na^+ out of the cell and 3 K^+ into the cell

(C) 3 Na^+ out of the cell and 2 K^+ into the cell

(D) Na^+ and K^+ in the same direction, down their gradients

2. What directly causes the sodium–potassium pump to change shape and release Na^+ outside the cell?

(A) Phosphorylation of the pump by ATP

(B) Binding of glucose

(C) Diffusion of water through the pump

(D) Opening of an aquaporin

3. Intestinal cells absorb glucose even when the glucose concentration inside the cell is higher than in the intestine. A drug that blocks the sodium–potassium pump would most likely

(A) increase glucose uptake, because more Na^+ would remain inside the cell

(B) have no effect, because glucose uptake does not involve sodium

(C) convert glucose transport into simple diffusion

(D) eventually reduce glucose uptake, because the Na^+ gradient that powers the cotransporter would dissipate

4. A white blood cell engulfs a bacterium and digests it. Which processes are involved, in order?

(A) Exocytosis, then fusion with a ribosome

(B) Phagocytosis, then fusion of the vesicle with a lysosome

(C) Pinocytosis, then osmosis

(D) Facilitated diffusion, then phagocytosis

5. Familial hypercholesterolemia is caused by defective LDL receptors. Which process is most directly impaired?

(A) Phagocytosis

(B) Pinocytosis

(C) Receptor-mediated endocytosis

(D) Exocytosis of insulin

PART 1 · THE LESSONS

2.10

TOPIC 2.10

Cell Compartmentalization

LEARNING TARGETS

- Explain how internal membranes allow different reactions to occur in different compartments at the same time.
- Explain how compartmentalization increases efficiency and protects the cell.
- Describe how membrane folding increases surface area for reactions.

Why Divide the Cell into Rooms?

A house has a kitchen, a bedroom, and a bathroom so that different activities can happen at once without getting in each other's way. Eukaryotic cells work the same way. Internal membranes divide the cell into **compartments**, each with its own chemical conditions. This **compartmentalization** has four main benefits:

1. **Separate, even opposite, reactions can run at the same time.** Fatty acids can be built in the cytosol while they are broken down in peroxisomes and mitochondria.
2. **Each compartment keeps its own ideal conditions.** Lysosomes keep a pH of about 5 for their enzymes, while the cytosol stays near pH 7.2.
3. **Enzymes and substrates are concentrated.** Keeping all the enzymes for one pathway in one small space speeds up reactions.
4. **Harmful substances are isolated.** Hydrolytic enzymes stay in lysosomes; hydrogen peroxide is made and destroyed inside peroxisomes.

Membranes as Work Surfaces

Membranes don't just separate compartments; they also hold the enzymes and transport proteins for many reactions. Membranes can also maintain **gradients** between compartments—for example, a high concentration of H^+ in the space between the two mitochondrial membranes—which cells use to make ATP (Unit 3). **Folding** (cristae, thylakoids, the rough ER) packs a large membrane area into a small volume, increasing the number of reactions that can occur.

COMPARTMENT	SPECIAL CONDITIONS	REACTIONS IT ALLOWS
Nucleus	DNA protected and separated from ribosomes	Transcription and RNA processing, separated from translation
Rough ER lumen	Folding enzymes; site for adding sugars	Folding and modification of proteins for secretion or membranes

COMPARTMENT	SPECIAL CONDITIONS	REACTIONS IT ALLOWS
Lysosome	Acidic pH (≈5); hydrolytic enzymes	Digestion without harming the rest of the cell
Mitochondrion	Two membranes; matrix and intermembrane space	Krebs cycle (matrix); electron transport and ATP synthesis (inner membrane)
Chloroplast	Thylakoid space and stroma	Light reactions (thylakoids); Calvin cycle (stroma)

◆ **MISCONCEPTION ALERT**

Prokaryotes are not "less evolved" because they lack compartments. Prokaryotes are extremely successful. Their small size gives them a high SA:V ratio, and some use infoldings of the plasma membrane for specialized reactions. Compartmentalization is one solution to the problem of organizing a cell, especially a large one.

■ **KEY TERMS**

Compartmentalization
Division of the cell interior into membrane-bound regions with different functions.

Intermembrane space
Region between the inner and outer membranes of a mitochondrion or chloroplast.

Check Your Understanding 2.10

1. Which is the best explanation of how compartmentalization benefits a eukaryotic cell?

(A) It allows the cell to grow without limit.
(B) It allows reactions that require different conditions to occur simultaneously in separate regions.
(C) It removes the need for a plasma membrane.
(D) It makes the cell permeable to all substances.

2. Peroxisomes produce hydrogen peroxide, which is toxic, and also contain catalase, which breaks it down. This arrangement best illustrates that compartmentalization

(A) isolates potentially harmful reactions from the rest of the cell
(B) increases the cell's SA:V ratio
(C) is found only in prokaryotes
(D) prevents all chemical reactions

3. In eukaryotes, transcription occurs in the nucleus and translation occurs in the cytoplasm. A direct consequence of this separation is that

(A) mRNA can be processed before it is translated
(B) proteins are made inside the nucleus
(C) DNA is translated directly into protein
(D) ribosomes are found only in the nucleus

4. Which feature of the chloroplast increases the surface area available for the light reactions?

(A) The stroma

(B) The cell wall

(C) The central vacuole

(D) Stacked thylakoid membranes

PART 1 · THE LESSONS

2.11

TOPIC 2.11

Origins of Cell Compartmentalization

LEARNING TARGETS

- Describe the endosymbiotic theory, step by step.
- Explain the evidence that mitochondria and chloroplasts were once free-living prokaryotes.
- Describe how the endomembrane system may have evolved by infolding of the plasma membrane.

The Endosymbiont Theory

Where did membrane-bound organelles come from? The **endosymbiont theory** (endosymbiotic theory) proposes that **mitochondria and chloroplasts were once free-living prokaryotes** that were engulfed by a larger host cell. Instead of being digested, they survived inside it, and the two cells came to depend on each other. An **endosymbiont** is an organism that lives inside the cells of another.

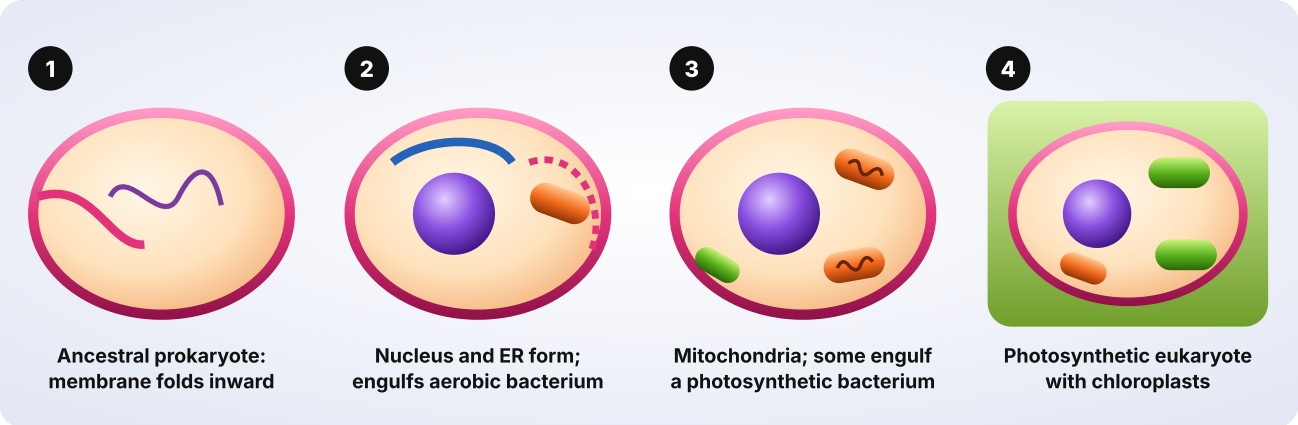


Figure 2.13 A simplified model of the origin of eukaryotic organelles. Infolding of the plasma membrane is thought to have produced the nuclear envelope and endomembrane system; endosymbiosis produced mitochondria and, later, chloroplasts.

Step by Step

1. **Infolding.** An ancestral prokaryotic cell's plasma membrane folded inward. These folds pinched off to form the **nuclear envelope** and the **endomembrane system** (ER and Golgi).
2. **First engulfment.** This early eukaryotic cell engulfed an **aerobic** (oxygen-using) prokaryote but did not digest it.
3. **Mutual benefit.** The engulfed cell provided ATP from aerobic respiration; the host provided protection and nutrients. Over many generations, the endosymbiont became the **mitochondrion**. All eukaryotes descend from this cell.

4. **Second engulfment.** Later, some of these cells engulfed a **photosynthetic** prokaryote (related to cyanobacteria), which became the **chloroplast**. These cells gave rise to algae and plants.

The Evidence

Mitochondria and chloroplasts share many features with bacteria that other organelles lack:

EVIDENCE	WHY IT SUPPORTS ENDOSYMBIOSIS
Double membrane	The inner membrane matches the original prokaryote's plasma membrane; the outer membrane came from the host's engulfing vesicle.
Own circular DNA	Like a bacterial chromosome, and unlike the linear chromosomes in the nucleus.
Own ribosomes, similar in size to bacterial ribosomes	They make some of the organelle's own proteins, and they are sensitive to some antibiotics that affect bacterial ribosomes.
Reproduce by a process like binary fission	They divide on their own inside the cell, the way bacteria do; the cell cannot make them from scratch.
Similar size to bacteria	Mitochondria are about the size of a typical bacterium.
DNA sequence comparisons	Mitochondrial DNA is most similar to that of a group of aerobic bacteria; chloroplast DNA is most similar to that of cyanobacteria.

● **EXAM TIP**

On a free-response question about endosymbiosis, you are often asked for **evidence**. Each piece of evidence earns credit only if you explain the comparison: not just "mitochondria have DNA," but "mitochondria have their own *circular* DNA, *like that of bacteria*, which suggests they were once independent prokaryotes."

◆ **MISCONCEPTION ALERT**

Mitochondria today cannot live on their own. Over time, many of the endosymbiont's genes moved to the host's nucleus, so mitochondria depend on proteins made by the cell. Also, the double membrane of the nucleus is explained by *infolding*, not endosymbiosis.

■ **KEY TERMS**

Endosymbiont theory
Theory that mitochondria and chloroplasts descend from prokaryotes engulfed by a host cell.

Endosymbiont
An organism that lives inside another organism's cells.

Binary fission
Prokaryotic cell division by splitting in two.

Infolding
Inward folding of the plasma membrane, thought to have formed the endomembrane system.

Check Your Understanding 2.11

1. Which observation provides support for the endosymbiont theory?

- **(A)** Mitochondria contain circular DNA similar to that of bacteria.
- **(B)** The Golgi complex has a *cis* and *trans* face.
- **(C)** Lysosomes contain hydrolytic enzymes.
- **(D)** The nucleus contains linear chromosomes.

2. According to the endosymbiont theory, the inner membrane of a mitochondrion was originally

- **(A)** the host cell's nuclear envelope
- **(B)** the plasma membrane of the engulfed prokaryote
- **(C)** part of the host cell's Golgi complex
- **(D)** the cell wall of the host

3. Which sequence of events is most consistent with the endosymbiont theory?

- **(A)** Chloroplasts arose first, then mitochondria, then the nucleus.
- **(B)** Mitochondria were acquired before chloroplasts, so all eukaryotes have mitochondria but only some have chloroplasts.
- **(C)** Mitochondria and chloroplasts arose by infolding of the plasma membrane.
- **(D)** The nucleus was engulfed as a free-living bacterium.

4. An antibiotic inhibits bacterial ribosomes but not ribosomes in the eukaryotic cytosol. In some patients it causes side effects in tissues with many mitochondria. Which explanation is most consistent with this?

- **(A)** Mitochondria have no ribosomes.
- **(B)** The antibiotic blocks the sodium–potassium pump.
- **(C)** Mitochondrial ribosomes resemble bacterial ribosomes, so the drug can also affect them.
- **(D)** The antibiotic dissolves the plasma membrane.

5. Which structure is thought to have arisen from infolding of the plasma membrane rather than from endosymbiosis?

- **(A)** Mitochondrion
- **(B)** Chloroplast
- **(C)** Ribosome
- **(D)** Endoplasmic reticulum

PART 1 · VISUAL REVIEW

Visual Review: Mind Maps

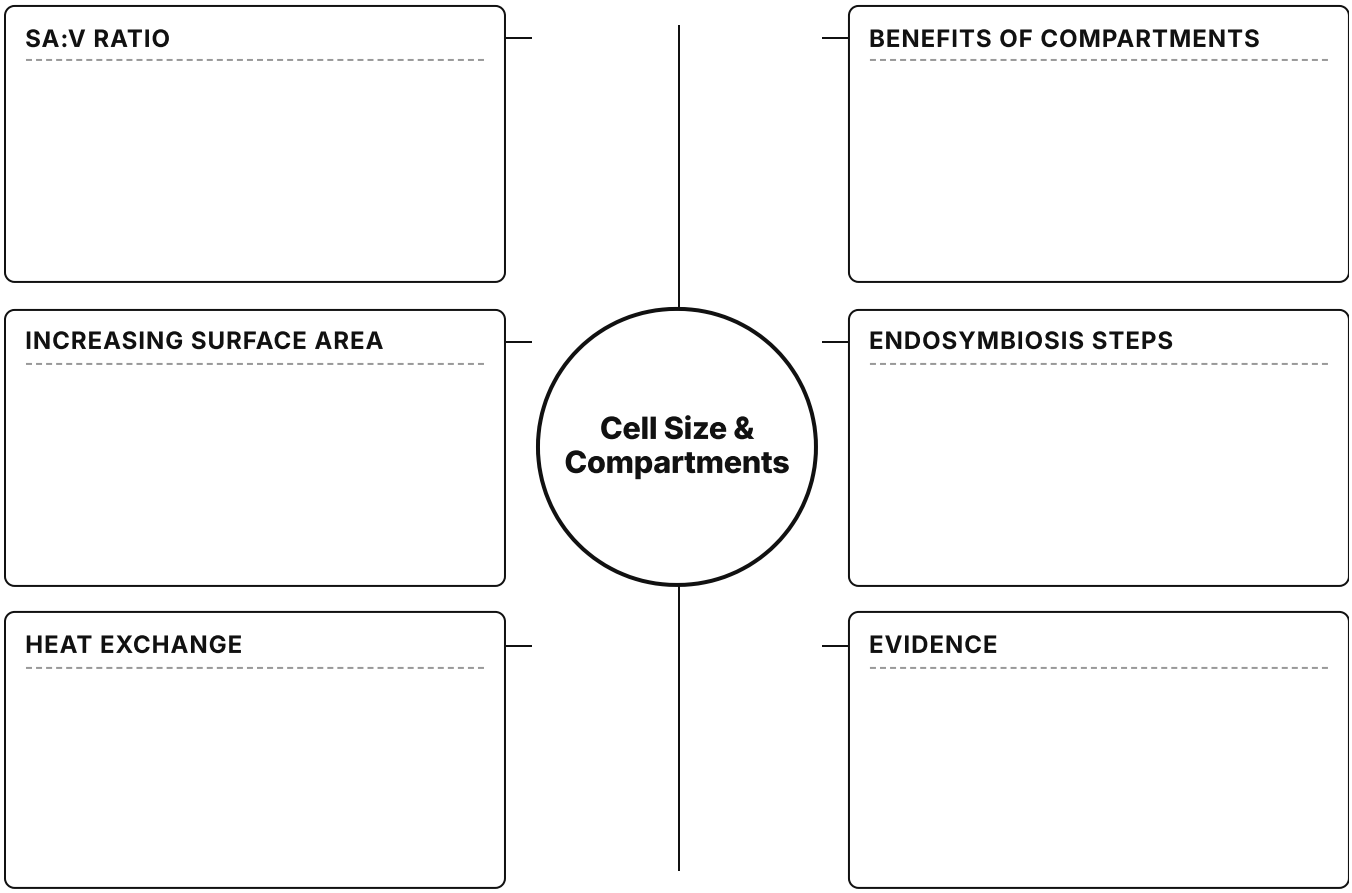
The next three pages summarize Unit 2 on single pages: the eukaryotic cell, the plasma membrane, and membrane transport. Each map puts one big idea at the center and branches out to the structures, processes, and examples that AP questions ask you to connect.

HOW TO STUDY WITH A MIND MAP

1. **Read the whole map once**, then trace each branch back to the center and say how it connects.
2. **Cover one branch** and recite everything under it. Uncover it and check.
3. **Redraw the map from memory** on blank paper. Compare and fill in what you missed in a different color.
4. **Find the links between maps.** Gradients, membranes, and ATP appear on every map. Why?

Your Turn: Build a Cell Size & Compartments Mind Map

There is no printed map for Topics 2.3, 2.10, and 2.11—you will make it. Fill in each box from memory, then check your work against the lessons.



Mind Map 1 · The Eukaryotic Cell

Eukaryotic Cell
structure → function
membrane-bound organelles

INFORMATION CENTER

- **Nucleus:** double membrane + pores; stores DNA; transcription
- **Nucleolus:** assembles ribosomal subunits
- **Ribosomes:** rRNA + protein; no membrane; make proteins; in *all* cells
- Free ribosomes → cytosol proteins; bound → secreted/membrane proteins

ENDOMEMBRANE SYSTEM

- **Rough ER:** ribosomes; folds and modifies proteins
- **Smooth ER:** lipids, detox, Ca^{2+} storage
- **Golgi:** modifies, sorts, ships (*cis* in, *trans* out)
- **Vesicles** link the parts
- Path: ER → vesicle → Golgi → vesicle → membrane

DIGESTION & STORAGE

- **Lysosome:** hydrolytic enzymes, pH ≈ 5; autophagy, apoptosis
- **Peroxisome:** breaks down fatty acids; catalase destroys H_2O_2
- **Central vacuole (plants):** water storage, turgor pressure
- **Contractile vacuole:** pumps out excess water

ENERGY ORGANELLES

- **Mitochondria:** double membrane; cristae ↑ surface area; matrix; ATP by cellular respiration
- **Chloroplasts:** double membrane; thylakoids (grana) + stroma; photosynthesis
- Both have own circular DNA and ribosomes (endosymbiosis)
- Plants have *both*

SUPPORT & MOVEMENT

- **Cytoskeleton:** microtubules, microfilaments (actin), intermediate filaments
- Shape, vesicle tracks, chromosome movement, cilia/flagella
- **Centrosome/centrioles** (animals): organize microtubules
- **Cell wall:** cellulose (plants), chitin (fungi), peptidoglycan (bacteria)

COMPARE CELL TYPES

- **Prokaryote:** no nucleus (nucleoid), no membrane organelles, ribosomes, small
- **Animal:** lysosomes, centrioles; no wall, no chloroplasts
- **Plant:** wall, chloroplasts, central vacuole + mitochondria

Link to Unit 3: mitochondria and chloroplasts are where cellular respiration and photosynthesis happen.

Mind Map 2 · The Plasma Membrane

PHOSPHOLIPID BILAYER

- Amphipathic phospholipids: hydrophilic heads, hydrophobic tails
- Heads face water on both sides; tails form the core
- Forms spontaneously (hydrophobic interactions)
- Hydrophobic core is the main barrier

MEMBRANE PROTEINS

- **Integral:** embedded; transmembrane span bilayer
- Nonpolar amino acids in core, polar/charged outside
- **Peripheral:** on the surface
- Jobs: transport, enzymes, receptors, recognition, junctions, attachment

CARBOHYDRATES

- **Glycoproteins** and **glycolipids** on the outer surface
- Cell–cell recognition: self vs. nonself, blood types

Plasma Membrane

fluid mosaic
selectively permeable

FLUIDITY

- ↑ temperature → more fluid
- Unsaturated (kinked) tails → more fluid
- Saturated (straight) tails → less fluid
- **Cholesterol** buffers: ↓ fluidity when warm, prevents packing when cold

PERMEABILITY

- **Free:** small nonpolar (O_2, CO_2, steroids)
- **Slow:** small polar (H_2O, glycerol)
- **Need proteins:** ions, large polar (glucose)
- **Need vesicles:** macromolecules

OUTSIDE THE MEMBRANE

- **Cell wall:** porous, structural, prevents bursting
- **ECM** (animals): collagen and other glycoproteins

Link to Unit 1: phospholipid structure (amphipathic) and R-group polarity explain the whole membrane.

Mind Map 3 · Membrane Transport

Membrane Transport

direction?
energy?
protein?

PASSIVE TRANSPORT

- Down the gradient (high → low); no energy
- **Simple diffusion:** through bilayer
- **Facilitated diffusion:** channels (aquaporins, gated ion channels) and carriers (GLUT)
- Facilitated rate saturates

ACTIVE TRANSPORT

- Against the gradient (low → high); requires energy
- **Na^+/K^+ pump:** ATP; 3 Na^+ out, 2 K^+ in
- Phosphorylation → shape change
- **Proton pumps** build H^+ gradients

OSMOSIS

- Diffusion of water across a membrane
- Water moves toward higher solute concentration
- **Hypotonic** → water in: animal lyses, plant turgid
- **Isotonic** → no net change: plant flaccid
- **Hypertonic** → water out: crenation, plasmolysis

GRADIENTS AS ENERGY

- **Membrane potential:** inside negative
- **Electrochemical gradient** = concentration + charge
- **Cotransport:** one substance down powers another up (Na^+–glucose; H^+–sucrose)

WATER POTENTIAL

- Water moves from high Ψ to low Ψ
- $\Psi = \Psi_P + \Psi_S$; $\Psi_S = -iCRT$
- Pure water Ψ = 0; solutes lower Ψ
- Osmoregulation: contractile vacuole, kidneys

BULK TRANSPORT

- **Exocytosis:** vesicle fuses, releases outside
- **Phagocytosis:** large particles
- **Pinocytosis:** fluid, nonspecific
- **Receptor-mediated:** specific (LDL)

Link to Units 3 and 4: H^+ gradients drive ATP synthase; ion gradients drive nerve signals.

PART 2 · SKILLS AND REVIEW

Science Practices and Unit 2 Math

The AP Biology Exam has two sections: multiple choice and free response. In recent years the free-response section has included two long questions (one of which involves interpreting an experiment) and four short questions that target specific skills such as experimental design, data analysis, models, and evolutionary trees. Always confirm the current format in the official College Board course description. Unit 2 is especially rich in calculation and lab-analysis questions, so this chapter focuses on them.

Know Your Task Verbs

VERB	WHAT IT ASKS FOR	UNIT 2 EXAMPLE OF A FULL-CREDIT RESPONSE
Identify	Name or state the answer. No explanation needed.	"Identify the organelle that modifies and packages proteins." → The Golgi complex.
Describe	Give the relevant characteristics.	"Describe the fluid mosaic model." → A phospholipid bilayer in which proteins, cholesterol, and carbohydrates are embedded or attached, and in which the components can move laterally.
Explain	Give the *how* or *why*; connect cause and effect.	"Explain why cells are small." → As a cell grows, volume increases faster than surface area, so SA:V decreases and the membrane cannot exchange materials fast enough to supply the cytoplasm.
Predict	State an expected outcome.	"Predict what happens to an elodea cell in 1 M NaCl." → It will lose water and plasmolyze.
Justify	Give evidence or reasoning that supports a claim.	"Justify your prediction." → The solution has a lower water potential than the cell, so water moves out of the cell by osmosis.
Calculate	Perform the calculation and show work.	"Calculate Ψ_S of 0.5 M sucrose at 20°C." → $-(1)(0.5)(0.0831)(293) = -12.2$ bars.
Construct / Represent	Draw a graph, table, or diagram.	Plot percent change in mass (y) against sucrose molarity (x), with labeled axes, units, and a line of best fit.
Support / Refute	Use evidence to argue for or against a claim.	"The claim that mitochondria were free-living is supported because they have circular DNA and bacteria-like ribosomes."

Unit 2 Calculation Toolkit

PERCENT CHANGE IN MASS

% change = (final mass − initial mass) ÷ initial mass × 100. **Example:** a potato core goes from 2.50 g to 2.20 g. (2.20 − 2.50) ÷ 2.50 × 100 = **−12%**. A negative value means the core lost water, so the solution was **hypertonic** to the potato. Using percent change (instead of raw mass) lets you compare cores that started at different masses.

FINDING THE ISOTONIC POINT FROM A GRAPH

1. Plot percent change in mass (y-axis) against solution molarity (x-axis).
2. Draw a line of best fit through the points.
3. Find where the line crosses **0% change**. Read the molarity on the x-axis.
4. That molarity is **isotonic** to the tissue; it estimates the tissue's solute concentration. Use it as C in $\Psi_S = -iCRT$ to estimate the tissue's solute potential.

SA:V QUICK REFERENCE

Cube: SA:V = $6/s$. Sphere: SA:V = $3/r$. When the side or radius **doubles**, SA:V is **halved**; surface area becomes 4 times larger but volume becomes 8 times larger.

Experimental Design Vocabulary

Independent variable
The factor the experimenter changes (x-axis). Example: sucrose molarity.

Dependent variable
The factor measured in response (y-axis). Example: % change in mass.

Controlled variables
Conditions kept the same: tissue type, core size, time, temperature.

Negative control
A group in which no response is expected (e.g., cores in distilled water show the effect of zero solute).

Positive control
A group in which a known response is expected; shows the method can detect an effect.

Null hypothesis
The prediction that the independent variable has no effect.

Error bars
Often $\pm 2\ SE_{\bar{x}}$ (about 95% confidence). Non-overlapping bars suggest a significant difference; heavily overlapping bars suggest no significant difference.

THE CLAIM–EVIDENCE–REASONING TEMPLATE

Claim: State your answer in one sentence. **Evidence:** Point to specific data. **Reasoning:** Explain the biological principle. Example: "*Claim:* The potato cells have a solute concentration of about 0.3 M. *Evidence:* Cores gained 5% mass in 0.2 M sucrose and lost 4% in 0.4 M; the line crosses 0% at about 0.3 M. *Reasoning:* At the isotonic concentration, water potential inside and outside the cells is equal, so there is no net osmosis and no change in mass."

Unit 2 Master Comparisons

Prokaryotic, Animal, and Plant Cells

STRUCTURE	PROKARYOTE	ANIMAL CELL	PLANT CELL
Plasma membrane	Yes	Yes	Yes
Ribosomes	Yes (smaller)	Yes	Yes
DNA	Circular, in nucleoid	Linear, in nucleus	Linear, in nucleus
Nucleus, ER, Golgi	No	Yes	Yes
Mitochondria	No	Yes	Yes
Chloroplasts	No (some have photosynthetic membranes)	No	Yes
Cell wall	Usually (peptidoglycan)	No	Yes (cellulose)
Central vacuole	No	No	Yes
Lysosomes / centrioles	No	Yes	Rare / No

Every Way Across the Membrane

MECHANISM	DIRECTION	ENERGY?	PROTEIN OR VESICLE?	EXAMPLES
Simple diffusion	High → low	No	Neither	O_2, CO_2, steroid hormones
Osmosis	High Ψ → low Ψ	No	Often aquaporins	Water
Facilitated diffusion (channel)	High → low	No	Channel protein	Ions through gated channels; water through aquaporins
Facilitated diffusion (carrier)	High → low	No	Carrier protein	Glucose through GLUT
Primary active transport	Low → high	Yes, ATP	Carrier (pump)	Na^+/K^+ pump; proton pump
Cotransport	One down, one up	Indirect (gradient made by a pump)	Carrier	Na^+–glucose; H^+–sucrose
Endocytosis	Into the cell	Yes	Vesicle	Phagocytosis, pinocytosis, receptor-mediated (LDL)
Exocytosis	Out of the cell	Yes	Vesicle	Secretion of insulin, neurotransmitters

Tonicity at a Glance

OUTSIDE SOLUTION	WATER POTENTIAL OUTSIDE	WATER MOVES	ANIMAL CELL	PLANT CELL
Hypotonic	Higher than cell	In	Lyses	Turgid (ideal)
Isotonic	Equal	No net	Normal (ideal)	Flaccid
Hypertonic	Lower than cell	Out	Crenated	Plasmolyzed

Unit 2 One-Page Review Sheet

2.1–2.2 Organelles

- All cells: membrane, cytoplasm, DNA, **ribosomes** (evidence of common ancestry).
- Prokaryote: nucleoid, no membrane organelles. Eukaryote: nucleus + organelles.
- Rough ER (proteins) → vesicle → Golgi (modify, sort) → vesicle → exocytosis.
- Smooth ER: lipids, detox, Ca^{2+}. Lysosome: hydrolysis at pH 5. Peroxisome: H_2O_2.
- Mitochondria: cristae ↑ area for ATP. Chloroplasts: thylakoids + stroma.
- Plants: wall, chloroplasts, central vacuole, *and* mitochondria.

2.3 Cell Size

- Volume $\propto L^3$, area $\propto L^2$ → bigger cell = **lower SA:V** = less efficient exchange.
- Increase area: microvilli, root hairs, folds, flattened shapes.
- Small animals: high SA:V, lose heat fast, high metabolic rate per gram.

2.4–2.5 Membranes

- Fluid mosaic: phospholipid bilayer + proteins + cholesterol + carbohydrates.
- Integral (transmembrane) vs. peripheral proteins.
- Glycoproteins/glycolipids: recognition.
- Fluidity ↑: heat, unsaturated tails. Cholesterol buffers.
- Small nonpolar cross freely; ions and large polar need proteins.
- Cell wall: porous, structural.

2.6–2.7 Passive Transport

- Passive = down gradient, no energy. Active = up gradient, energy.
- Facilitated diffusion: channels (aquaporins, gated ion channels), carriers (GLUT). Rate **saturates**.

2.8 Tonicity & Water Potential

- Water moves toward higher solute concentration = from high Ψ to low Ψ.
- Hypotonic: water in. Hypertonic: water out. Isotonic: no net.
- $\Psi = \Psi_P + \Psi_S$; $\Psi_S = -iCRT$; $R = 0.0831$; T in K; i = 1 sucrose, 2 NaCl.
- −2 bars is *higher* than −5 bars.
- Osmoregulation: contractile vacuoles, kidneys.

2.9 Mechanisms

- Na^+/K^+ pump: ATP, 3 Na^+ out, 2 K^+ in → inside negative.
- Electrochemical gradient = concentration + charge.
- Cotransport: Na^+–glucose, H^+–sucrose.
- Endocytosis: phago-, pino-, receptor-mediated. Exocytosis: secretion.

2.10–2.11 Compartments

- Separate conditions, simultaneous reactions, concentrated enzymes, isolated hazards.
- Endosymbiosis: mitochondria then chloroplasts. Evidence: double membrane, circular DNA, own bacteria-like ribosomes, binary fission, DNA similarity.
- Nucleus and ER: infolding of plasma membrane.

PART 3 · PRACTICE TEST

Section I: Multiple Choice

Time suggestion: 35 minutes · 25 questions. Choose the one best answer for each question. Mark your answers, then check them in Part 4. Record your score in the Score Tracker at the end of the book. (R = 0.0831 L·bar/mol·K.)

1. Which structure is found in every living cell?

(A) Nucleus
(B) Mitochondrion
(C) Cell wall
(D) Ribosome

2. Which organelle contains its own DNA and ribosomes?

(A) Mitochondrion
(B) Golgi complex
(C) Lysosome
(D) Smooth ER

3. Researchers briefly expose pancreatic cells to radioactive amino acids and then track where the radioactivity appears over time. In what order would the radioactivity most likely appear?

(A) Golgi → rough ER → secretory vesicles
(B) Nucleus → rough ER → lysosomes
(C) Rough ER → Golgi → secretory vesicles
(D) Secretory vesicles → Golgi → rough ER

4. Cells in the testes that produce the steroid hormone testosterone would be expected to have an extensive

(A) rough ER
(B) smooth ER
(C) central vacuole
(D) chloroplast network

5. What is the surface area-to-volume ratio of a cube with sides of 3 cm?

(A) 1 : 1
(B) 3 : 1
(C) 2 : 1
(D) 6 : 1

6. Three cells have the dimensions shown. Which cell would exchange materials with its environment most efficiently?

CELL	SURFACE AREA (MM^2)	VOLUME (MM^3)
W	96	32
X	150	125
Y	216	216
Z	384	512

(A) W
(B) X
(C) Y
(D) Z

7. The portion of a transmembrane protein that is embedded within the phospholipid bilayer is most likely to contain amino acids that are

(A) positively charged
(B) negatively charged
(C) polar
(D) nonpolar

8. A patient receives a blood transfusion of the wrong type, and the immune system attacks the donated red blood cells. Which membrane components does the immune system most directly recognize?

(A) Cholesterol molecules
(B) Carbohydrate chains of glycoproteins and glycolipids
(C) Fatty acid tails
(D) Aquaporins

9. Winter wheat plants exposed to cold temperatures increase the proportion of unsaturated fatty acids in their membranes. This change most likely helps the plants by

(A) keeping membranes fluid at low temperatures
(B) making membranes less permeable to water
(C) increasing the number of chloroplasts
(D) preventing the membrane from forming a bilayer

10. Which substance would diffuse across a phospholipid bilayer most rapidly without the help of a protein?

(A) Cl^-
(B) Sucrose
(C) CO_2
(D) A protein hormone

11. The rate of uptake of substance S into cells increases with external concentration until it reaches a maximum, after which further increases in concentration have no effect. ATP inhibitors do not change the rate. Substance S most likely enters by

(A) simple diffusion
(B) facilitated diffusion through a carrier protein
(C) primary active transport
(D) phagocytosis

12. What is the solute potential of a 0.10 M NaCl solution at 27°C? (Assume NaCl fully ionizes.)

(A) −2.49 bars
(B) +4.99 bars
(C) −0.50 bars
(D) −4.99 bars

13. A plant cell with a water potential of −0.5 MPa is placed in pure water. Which is the most likely result?

(A) Water enters the cell until its pressure potential increases enough that its water potential reaches 0 MPa.
(B) Water leaves the cell until it plasmolyzes.
(C) The cell bursts because plant cells cannot resist water pressure.
(D) There is no net movement because pure water has a water potential of −0.5 MPa.

14. Leaf cells of the aquatic plant *Elodea* are observed under a microscope after a drop of concentrated salt solution is added. The cell membranes are seen pulling away from the cell walls. This is because

(A) salt enters the cells by osmosis
(B) the cell walls shrink in salt solution
(C) water moves out of the cells toward the lower water potential outside
(D) water moves into the central vacuoles

15. Cyanide stops ATP production in animal cells. Over time, how would cyanide affect the ion concentrations inside a cell?

(A) Na^+ would decrease and K^+ would increase inside the cell.
(B) Na^+ would increase and K^+ would decrease inside the cell, as ions leak down their gradients.
(C) Ion concentrations would not change because ions move only by diffusion.
(D) Both Na^+ and K^+ would increase inside the cell.

16. In plant cells, sucrose is loaded into phloem cells through an H^+/sucrose cotransporter. A drug that inhibits the plasma membrane proton pump would most likely

(A) increase sucrose uptake by making the cell more acidic
(B) have no effect, because sucrose moves by simple diffusion
(C) increase the number of chloroplasts
(D) decrease sucrose uptake, because the H^+ gradient that powers the cotransporter would decline

17. Cells take up cholesterol by binding LDL particles to specific receptors in coated pits on the membrane. This process is

(A) receptor-mediated endocytosis
(B) pinocytosis
(C) exocytosis
(D) facilitated diffusion

18. A drug raises the pH inside lysosomes from about 5 to about 7. Which is the most likely effect on the cell?

(A) Protein synthesis on the rough ER will increase.
(B) The cell will make more ATP.
(C) Worn-out organelles and engulfed material will accumulate because lysosomal enzymes become less active.
(D) The plasma membrane will become impermeable to water.

19. Which observation provides the strongest evidence that chloroplasts descended from free-living cyanobacteria?

(A) Chloroplasts are green.
(B) Chloroplast DNA is circular and its sequences are most similar to those of cyanobacteria.
(C) Chloroplasts are found only in plants.
(D) Chloroplasts are surrounded by the endoplasmic reticulum.

20. Within a single eukaryotic cell, fatty acids are synthesized in the cytosol while other fatty acids are broken down inside peroxisomes. This is possible primarily because

(A) the two reactions use the same enzymes
(B) prokaryotes lack peroxisomes
(C) the cytosol has no enzymes
(D) membranes separate the cell into compartments with different enzymes and conditions

21. A dialysis-tubing bag (permeable to water but not sucrose) containing 20% sucrose is placed in a beaker of 5% sucrose. After one hour, the bag will most likely have

(A) gained mass, because water moved into the bag
(B) lost mass, because sucrose moved out of the bag
(C) lost mass, because water moved out of the bag
(D) stayed the same mass, because the bag is impermeable to sucrose

22. Sweet potato cores were soaked in sucrose solutions. Based on the data, which is the best estimate of the solute concentration of the sweet potato cells?

SUCROSE (M)	0.2	0.4	0.6	0.8
% change in mass	+11	+4	−3	−10

(A) 0.3 M
(B) 0.4 M
(C) 0.5 M
(D) 0.7 M

23. A freshwater *Paramecium* is moved into water with progressively higher salt concentrations (still hypotonic to the cell). How would the rate at which its contractile vacuole expels water most likely change?

(A) It would increase, because more water would enter the cell.

(B) It would decrease, because less water would enter the cell by osmosis.

(C) It would stay the same, because the vacuole works at a fixed rate.

(D) It would stop immediately, because salt destroys contractile vacuoles.

24. A cell has a plasma membrane covered with thousands of microvilli and is packed with mitochondria near the membrane. The cell is most likely specialized for

(A) storing starch

(B) photosynthesis

(C) producing steroid hormones

(D) active absorption of nutrients

25. Frog egg cells normally have very few aquaporins and swell only slowly in a hypotonic solution. Eggs injected with mRNA for an aquaporin protein are placed in the same hypotonic solution. Compared with uninjected eggs, the injected eggs will most likely

(A) swell much faster and may burst, because water enters more rapidly through aquaporins

(B) shrink, because aquaporins pump water out of the cell

(C) show no change, because aquaporins transport only ions

(D) swell more slowly, because aquaporins block water movement

Section II: Free Response

Time suggestion: 40 minutes · 3 questions. Write your answers in complete sentences on the lines provided or on separate paper. Label each part. Scoring guidelines appear in Part 4.

Question 1 (Long Response, 9 points)

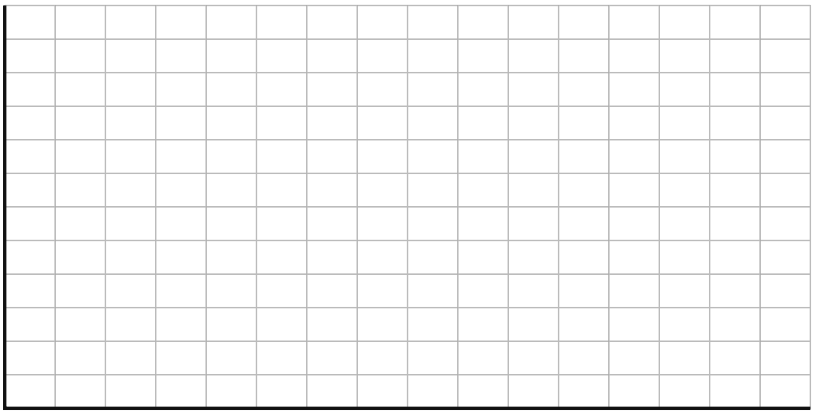
A student cut identical cylindrical cores from a single potato, recorded each core's mass, and placed groups of cores in sucrose solutions of different concentrations at 22°C. After 24 hours the cores were blotted dry and reweighed. The mean percent change in mass is shown below (mean ± 2SE$_{\bar{x}}$; n = 8 cores per solution).

SUCROSE (M)	0.0	0.2	0.4	0.6	0.8	1.0
% change in mass	+18.0 ± 1.6	+6.5 ± 1.2	−4.2 ± 1.0	−13.8 ± 1.4	−20.5 ± 1.5	−24.1 ± 1.8

(a) Describe osmosis. **Explain** why the cores in 0.0 M sucrose gained mass. **[2 pts]**

(b) Identify the independent variable and the dependent variable. **Explain** why the student recorded *percent* change in mass rather than the change in grams. **[2 pts]**

(c) Construct an appropriately labeled graph of the data, including error bars. **[2 pts]**



(d) Using the data, **estimate** the sucrose concentration that is isotonic to the potato cells. Then **calculate** the solute potential of the potato cells at 22°C, assuming their solute concentration equals this isotonic value. Show your work. **[2 pts]**

(e) The student repeats the experiment with cores from a sweet potato, whose cells contain a higher concentration of dissolved sugars. **Predict** how the isotonic concentration for sweet potato would compare with that of the potato, and **justify** your prediction. **[1 pt]**

Question 2 (Short Response, 4 points)

A teacher uses three cube-shaped models to represent cells of different sizes: 1 µm, 2 µm, and 4 µm on each side.

(a) Calculate the surface area-to-volume ratio of the 4 µm model. Show your work. **[1 pt]**

(b) Explain why the 1 µm cell would be more efficient than the 4 µm cell at obtaining nutrients and removing wastes. **[1 pt]**

(c) Describe one structural feature that allows a large cell to increase its surface area without greatly increasing its volume, and **identify** a cell type that has this feature. **[1 pt]**

(d) A mouse and an elephant are both endotherms. **Predict** which animal has the higher metabolic rate per gram of body mass, and **justify** your prediction using surface area-to-volume ratio. **[1 pt]**

Question 3 (Short Response, 4 points)

Intestinal epithelial cells absorb glucose through a sodium–glucose cotransporter, even when glucose is more concentrated inside the cell than in the intestine. Researchers measured glucose uptake under three conditions. Ouabain is a drug that specifically inhibits the sodium–potassium pump.

CONDITION	GLUCOSE UPTAKE (% OF CONTROL)
Control (normal Na^+ outside)	100
Ouabain added (after 30 min)	22
No Na^+ in outside solution	8

(a) Describe how the sodium–potassium pump establishes a Na^+ concentration gradient across the plasma membrane. **[1 pt]**

(b) Identify the type of transport by which glucose enters the intestinal cells, and **explain** the result for the "No Na^+" condition. **[1 pt]**

(c) Explain why ouabain reduced glucose uptake even though ouabain does not bind to the glucose cotransporter. **[1 pt]**

(d) The researchers measured uptake 2 minutes after adding ouabain and found it was 95% of control. **Explain** why the effect of ouabain was much smaller at 2 minutes than at 30 minutes. **[1 pt]**

PART 3 · EXTRA PRACTICE

Who Am I? Vocabulary Challenge

Precise vocabulary earns points on the AP Biology Exam. Each clue below describes an organelle, structure, molecule, or process from Unit 2. Write the term that best fits each clue. Answers are in Part 4.

HOW TO USE THIS SECTION

Work one section at a time. Mark any clue you had to guess, then reread that part of the lessons before checking the answers. Return to the marked clues a few days later.

Section A · Organelles and Cell Types

1. I am the double-membrane structure that stores a eukaryotic cell's DNA. Who am I? _____

2. I am the dense region inside the nucleus where ribosomal subunits are assembled. What am I?

_____

3. I am the openings in the nuclear envelope that let mRNA leave and proteins enter. What am I?

_____

4. I am built of rRNA and protein, have no membrane, and am found in every living cell. What am I?

_____

5. I am a network of membrane sacs studded with ribosomes, where proteins for secretion are made. What am I?

_____

6. I am a network of membrane tubules without ribosomes that makes lipids and detoxifies drugs. What am I?

_____

7. I am a stack of flattened sacs with a *cis* face and a *trans* face that modifies, sorts, and ships proteins. What am I?

_____

8. I am a small membrane sac that carries materials between parts of the endomembrane system. What am I?

_____

9. I am a membrane sac of hydrolytic enzymes that work best at about pH 5. What am I? _____

10. I am the process in which a lysosome digests a cell's own worn-out organelles. What am I? _____

11. I am programmed cell death, which lysosomes can help carry out. What am I? _____

12. I am the organelle that produces most of a eukaryotic cell's ATP through cellular respiration. What am I?

_____

13. I am the folds of the inner mitochondrial membrane that increase its surface area. What am I?

_____

14. I am the fluid inside the inner mitochondrial membrane. What am I? _____

15. I am the organelle that carries out photosynthesis in plants and algae. What am I? _____

16. I am the stacked membrane sacs inside a chloroplast where the light reactions occur. What am I?

_____

17. I am the fluid around the thylakoids where the Calvin cycle occurs. What am I? _____

18. I am the large, water-filled sac that creates turgor pressure in plant cells. What am I? _____

19. I am the membrane surrounding the central vacuole. What am I? _____

20. I am a single-membrane organelle that uses catalase to break down hydrogen peroxide. What am I?

_____

21. I am the network of protein fibers that gives a cell its shape and moves vesicles and chromosomes. What am I?

_____

22. I am the hollow tubes of tubulin that form cilia, flagella, and the spindle. What am I? _____

23. I am the thin actin fibers involved in muscle contraction and changes in cell shape. What am I?

_____

24. I am a pair of structures made of microtubule triplets, found in the centrosome of animal cells. What am I?

_____

25. I am the rigid layer outside the plasma membrane of plant cells, made of cellulose. What am I?

_____

26. I am the channels through plant cell walls that connect the cytoplasm of neighboring cells. What am I?

_____

27. I am the region of a prokaryotic cell that contains its DNA but has no membrane around it. What am I?

_____

28. I am a small, circular piece of extra DNA often found in bacteria. What am I? _____

29. I am the type of cell that has a nucleus and membrane-bound organelles. What am I? _____

30. I am the group of organelles—nuclear envelope, ER, Golgi, lysosomes, vacuoles, vesicles, and plasma membrane—that work together to make and ship proteins and lipids. What am I? _____

Section B · Membrane Structure and Permeability

31. I am the model that describes the membrane as a fluid bilayer with a patchwork of proteins and other molecules. What am I? _____

32. I am the term for a molecule, such as a phospholipid, with both hydrophilic and hydrophobic regions. What am I?

_____

33. I am the part of a phospholipid that faces the watery fluid inside and outside the cell. What am I?

_____

34. I am the part of a phospholipid that faces the interior of the bilayer, away from water. What am I?

_____

35. I am a membrane protein embedded in the bilayer. What type of protein am I? _____

36. I am an integral protein that spans the entire membrane from one side to the other. What am I?

_____

37. I am a protein loosely attached to the surface of the membrane, not embedded in it. What am I?

_____

38. I am a protein with a carbohydrate chain attached, used for cell–cell recognition. What am I? _____

39. I am a lipid with a carbohydrate chain attached, found on the outer face of the membrane. What am I?

_____

40. I am a steroid in animal membranes that buffers fluidity at high and low temperatures. What am I?

_____

41. I am the type of fatty acid tail whose double bonds create kinks that increase membrane fluidity. What am I?

_____

42. I am the type of fatty acid tail that is straight and packs tightly, decreasing fluidity. What am I?

_____

43. I am the property of a membrane that lets some substances cross more easily than others. What am I? _____

44. I am a small, nonpolar gas that diffuses directly through the bilayer and is used in cellular respiration. What am I? _____

45. I am the region of the membrane that blocks ions and large polar molecules. What am I? _____

46. I am a channel protein that allows water to cross the membrane rapidly. What am I? _____

47. I am the network of glycoproteins such as collagen outside animal cells. What am I? _____

48. I am the polysaccharide that makes up plant cell walls. What am I? _____

49. I am the polysaccharide that makes up fungal cell walls. What am I? _____

50. I am the material that makes up most bacterial cell walls. What am I? _____

51. I am the kind of hormone that can cross the membrane without a transport protein because I am a nonpolar lipid. What am I? _____

52. I am the general term for membrane proteins that help polar molecules and ions cross. What am I? _____

53. I am a membrane protein that binds a signal molecule and starts a response inside the cell. What am I? _____

54. I am the sideways movement of phospholipids within their layer that makes the membrane "fluid." What am I? _____

55. I am the arrangement of two layers of phospholipids with tails facing inward. What am I? _____

Section C · Transport, Tonicity, and Water Potential

56. I am a difference in the concentration of a substance between two regions. What am I? _____

57. I am the net movement of particles from higher to lower concentration. What am I? _____

58. I am the state in which molecules still move but there is no net change in concentration. What am I? _____

59. I am movement across a membrane down a gradient with no energy input from the cell. What am I? _____

60. I am passive movement through a channel or carrier protein. What am I? _____

61. I am a protein that binds a substance and changes shape to move it across the membrane. What am I? _____

62. I am a channel that opens or closes in response to a stimulus such as voltage or a signal molecule. What am I? _____

63. I am the condition in which all carrier proteins are busy, so the rate of transport stops increasing. What am I? _____

64. I am movement against a concentration gradient that requires energy. What am I? _____

65. I am the pump that uses ATP to move 3 Na^+ out of and 2 K^+ into an animal cell. What am I? _____

66. I am the addition of a phosphate group from ATP that changes the shape of a pump. What am I? _____

67. I am the voltage across a membrane caused by an unequal distribution of ions. What am I? _____

68. I am the combined effect of an ion's concentration gradient and the charge difference across a membrane. What am I? _____

69. I am transport in which one substance moving down its gradient powers another substance moving up its gradient. What am I? _____

70. I am an active transporter that pumps H^+ out of plant cells. What am I? _____

71. I am the diffusion of water across a selectively permeable membrane. What am I? _____

72. I am a solution with a higher solute concentration than the cell. What am I? _____

73. I am a solution with a lower solute concentration than the cell. What am I? _____

74. I am a solution with the same solute concentration as the cell. What am I? _____

75. I am the shrinking of a plant cell's cytoplasm away from its wall in a hypertonic solution. What am I?

_____

76. I am the firm condition of a plant cell in a hypotonic solution. What am I? _____

77. I am the measure of water's tendency to move; water moves from high to low values of me. What am I?

_____

78. I am the component of water potential caused by dissolved particles; I equal $-iCRT$. What am I?

_____

79. I am the organelle that pumps excess water out of freshwater protists. What am I? _____

80. I am the control of water balance and solute concentration in an organism. What am I? _____

Section D · Bulk Transport, Cell Size, and Compartments

81. I am the release of vesicle contents outside the cell when the vesicle fuses with the plasma membrane. What am I?

_____

82. I am the general process of taking material into the cell by folding in the plasma membrane. What am I?

_____

83. I am "cell eating": a cell engulfing a large particle such as a bacterium. What am I? _____

84. I am "cell drinking": nonspecific uptake of extracellular fluid in small vesicles. What am I? _____

85. I am endocytosis in which specific molecules, such as LDL, bind receptors before being taken in. What am I?

_____

86. I am the ratio that decreases as a cell grows larger, making exchange less efficient. What am I?

_____

87. I am the finger-like projections of intestinal cell membranes that increase surface area for absorption. What am I?

_____

88. I am the long extensions of plant root cells that increase the surface area for absorbing water. What am I?

_____

89. I am the SA:V ratio of a cube with sides of 2 units. What am I? _____

90. I am the division of a eukaryotic cell into membrane-bound regions with different conditions. What am I?

_____

91. I am the region between the inner and outer membranes of a mitochondrion. What am I? _____

92. I am the theory that mitochondria and chloroplasts descended from engulfed prokaryotes. What am I?

_____

93. I am an organism that lives inside the cells of another organism. What am I? _____

94. I am the kind of prokaryote that, once engulfed, became the mitochondrion. What type was I?

_____

95. I am the group of photosynthetic bacteria most closely related to chloroplasts. What am I? _____

96. I am the shape of the DNA found in mitochondria, chloroplasts, and bacteria. What am I? _____

97. I am the way mitochondria and chloroplasts reproduce, similar to bacterial cell division. What am I?

_____

98. I am the process, different from endosymbiosis, thought to have formed the nuclear envelope and ER. What am I?

_____

99. I am the type of cell that lacks a nucleus and membrane-bound organelles. What am I? _____

100. I am the enzyme in peroxisomes that converts hydrogen peroxide to water and oxygen. What am I?

_____

PART 4 · ANSWERS AND REFERENCE

Answers: Check Your Understanding

Topic 2.1

1. (C) Ribosomes are in all cells. Prokaryotes lack a nucleus (A), mitochondria (B), and a Golgi complex (D).

2. (C) A wall plus a central vacuole means a plant cell; root cells are underground and lack chloroplasts. Bacteria (A) have no nucleus; animal cells (B, D) have no wall.

3. (A) Smooth ER makes lipids and detoxifies. The Golgi does not make ATP (B), lysosomes digest (C), and the nucleolus assembles ribosomes (D).

4. (B) Lysosomal hydrolytic enzymes digest engulfed material and worn-out organelles.

5. (B) Ribosomes made of rRNA and protein are universal, reflecting common ancestry. Prokaryotes have no nucleus (A) or chloroplasts (C), and the cell types differ in size (D).

6. (D) Chloroplasts make sugar; mitochondria break it down to make ATP. Plants need both.

Topic 2.2

1. (B) Secreted proteins go rough ER → transport vesicle → Golgi → secretory vesicle → plasma membrane.

2. (A) Making and secreting large amounts of protein requires extensive rough ER and Golgi.

3. (D) Cristae increase the inner membrane area that holds electron transport proteins and ATP synthase.

4. (A) Enzymes that leak into the near-neutral cytosol work poorly, protecting the cell from self-digestion.

5. (B) Vesicles from the ER cannot deliver their proteins to the Golgi, so those proteins are not processed or secreted. Free-ribosome proteins (A) were never in this pathway.

6. (A) Smooth ER detoxifies drugs and poisons, especially in liver cells.

Topic 2.3

1. (B) SA = 6 × 25 = 150 µm^2; V = 125 µm^3; 150 ÷ 125 = 1.2. (Shortcut: $6/s = 6/5$.)

2. (C) Volume grows with the cube of length, surface area with the square, so SA:V falls. Surface area still increases (A is false).

3. (B) Microvilli add membrane area for absorption with little added volume.

4. (D) The smallest animal has the highest SA:V, loses heat fastest, and must produce the most heat per gram.

5. (C) A long, thin shape has the most surface per unit volume; a sphere has the least.

6. (A) The smallest cube has the highest SA:V and the shortest distance to its center, so the greatest fraction of its volume is reached in the same time.

Topic 2.4

1. (B) Hydrophilic heads face water; hydrophobic tails cluster away from it. No covalent bonds (C) or ATP (D) are involved.

2. (C) The bilayer interior is hydrophobic, so nonpolar R groups interact favorably with it.

3. (A) Carbohydrate chains on glycoproteins and glycolipids serve as identification tags.

4. (B) Kinked unsaturated tails keep membranes fluid in the cold.

5. (D) Cholesterol is a fluidity buffer. It is not a channel (B) or receptor (C).

Topic 2.5

1. (C) O_2 is small and nonpolar. Ions (A), glucose (B), and amino acids (D) need proteins.

2. (B) Steroids are lipids that dissolve in the hydrophobic core.

3. (A) Charged ions are repelled by the nonpolar interior. Size (B) is not the main issue; Na^+ is tiny.

4. (C) The wall is porous and structural. The plasma membrane, not the wall, controls entry (A, D).

5. (A) Without carrier proteins such as GLUT, glucose crosses a bare bilayer very slowly.

Topic 2.6

1. (C) Only active transport moves substances against their gradient with energy input. Facilitated diffusion also uses proteins (A).

2. (C) K^+ is already more concentrated inside, so moving more in is against the gradient. A channel (D) would let K^+ flow *out*.

3. (B) At dynamic equilibrium, movement continues in both directions with no net change.

4. (A) Vesicles fuse with the membrane and release their contents outside.

5. (D) An uneven distribution can do work as it evens out, like water behind a dam.

Topic 2.7

1. (B) Facilitated diffusion is passive, protein-assisted, and specific.

2. (A) Aquaporins allow rapid water movement.

3. (B) A plateau indicates saturated carrier proteins; simple diffusion (A) does not level off.

4. (C) Open channels let ions flow down their electrochemical gradient; no ATP is used.

5. (A) The maximum rate depends on the number of carriers. ATP (D) is not used in facilitated diffusion.

Topic 2.8

1. (C) Distilled water is hypotonic; water enters and the cell, lacking a wall, may lyse.

2. (A) A hypertonic solution draws water out; the membrane pulls away from the wall.

3. (B) $\Psi_S = -(1)(0.30)(0.0831)(300) = -7.48$ bars. Glucose does not ionize, so $i = 1$.

4. (D) −0.3 MPa is higher than −0.8 MPa, so water moves from the solution into the cell.

5. (B) Fresh water is hypotonic; without the contractile vacuole, water accumulates.

6. (C) The change in mass switches from gain to loss between 0.2 M and 0.4 M, so the isotonic point is about 0.3 M.

Topic 2.9

1. (C) Three Na^+ out, two K^+ in, both against their gradients.

2. (A) Phosphorylation by ATP changes the pump's shape so it opens to the outside.

3. (D) The cotransporter depends on the Na^+ gradient maintained by the pump. Once the gradient runs down, glucose uptake against its gradient fails.

4. (B) Phagocytosis forms a vesicle that fuses with a lysosome for digestion.

5. (C) LDL enters by receptor-mediated endocytosis; defective receptors block uptake.

Topic 2.10

1. (B) Compartments let reactions with different requirements happen at the same time.

2. (A) Making and destroying H_2O_2 in one compartment protects the rest of the cell.

3. (A) Because mRNA is made in the nucleus and translated outside it, it can be processed before translation.

4. (D) Stacked thylakoids increase the membrane area for the light reactions.

Topic 2.11

1. (A) Circular DNA like that of bacteria supports a prokaryotic origin. The other features have nothing to do with endosymbiosis.

2. (B) The inner membrane came from the engulfed cell; the outer came from the host.

3. (B) All eukaryotes have mitochondria (or descended from cells that had them), but only some have chloroplasts, so mitochondria came first.

4. (C) Mitochondrial ribosomes resemble bacterial ribosomes, so drugs that target bacterial ribosomes can affect them.

5. (D) The ER and nuclear envelope are thought to have formed by infolding; mitochondria and chloroplasts by endosymbiosis.

Answers: Practice Test

Section I: Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13
D	A	C	B	C	A	D	B	A	C	B	D	A

14	15	16	17	18	19	20	21	22	23	24	25
C	B	D	A	C	B	D	A	C	B	D	A

Section I: Explanations

1. (D) Ribosomes are universal. Prokaryotes lack nuclei and mitochondria; animal cells lack walls. (Topic 2.1)

2. (A) Mitochondria (and chloroplasts) have their own DNA and ribosomes, consistent with endosymbiosis. (2.1, 2.11)

3. (C) Proteins for secretion are made on the rough ER, processed in the Golgi, and packaged into secretory vesicles. (2.2)

4. (B) Steroids are lipids, made by the smooth ER. (2.2)

5. (C) SA = $6 \times 9 = 54$ cm^2; V = 27 cm^3; $54 \div 27 = 2$. (2.3)

6. (A) SA:V: W = 3.0, X = 1.2, Y = 1.0, Z = 0.75. The highest ratio is the most efficient. (2.3)

7. (D) Nonpolar R groups interact with the hydrophobic interior of the bilayer. (2.4)

8. (B) Blood types are determined by carbohydrate chains on the cell surface. (2.4)

9. (A) Unsaturated, kinked tails prevent tight packing and keep membranes fluid in the cold. (2.4)

10. (C) CO_2 is small and nonpolar. Ions, sugars, and proteins cannot cross a bare bilayer easily. (2.5)

11. (B) Saturation indicates a carrier; no ATP dependence means it is passive. Simple diffusion (A) does not saturate. (2.7)

12. (D) $\Psi_S = -(2)(0.10)(0.0831)(300) = -4.99$ bars. Solute potential is never positive (B). (2.8)

13. (A) Pure water ($\Psi = 0$) is higher than the cell, so water enters; the wall pushes back, raising Ψ_P until equilibrium. The wall prevents bursting (C). (2.8)

14. (C) The salt solution has a lower water potential, so water leaves and the cells plasmolyze. Salt does not move by osmosis (A). (2.8)

15. (B) Without ATP the pump stops, and Na^+ leaks in and K^+ leaks out down their gradients. (2.9)

16. (D) The proton pump builds the H^+ gradient that powers H^+/sucrose cotransport. (2.9)

17. (A) Binding of specific molecules to receptors defines receptor-mediated endocytosis. (2.9)

18. (C) Lysosomal enzymes need an acidic pH; at pH 7 they work poorly and undigested material builds up. (2.2, 2.10)

19. (B) Circular DNA plus sequence similarity to cyanobacteria is direct evidence of ancestry. (2.11)

20. (D) Compartmentalization separates opposing pathways. (2.10)

21. (A) Water moves into the bag toward the higher solute concentration; sucrose cannot leave (B). (2.8)

22. (C) Mass change crosses 0% between 0.4 M (+4%) and 0.6 M (−3%), at about 0.5 M. (2.8, data analysis)

23. (B) A smaller difference in solute concentration means less water enters, so less must be pumped out. (2.8)

24. (D) Microvilli add surface area; mitochondria supply ATP for active transport. (2.2, 2.3, 2.9)

25. (A) Aquaporins greatly increase water permeability, so the eggs swell rapidly in a hypotonic solution. (2.7, 2.8)

Section II: Scoring Guidelines

Question 1 (9 points)

(a) 2 points

- 1 pt — Describes osmosis: the diffusion (net movement) of water across a selectively permeable membrane from an area of lower solute concentration (higher water potential) to higher solute concentration (lower water potential).
- 1 pt — Explains the gain: 0.0 M sucrose (distilled water) is hypotonic to the potato cells (has a higher water potential), so water moved into the cells by osmosis.

(b) 2 points

- 1 pt — Independent variable: sucrose concentration (molarity). Dependent variable: percent change in mass. (Both required.)
- 1 pt — Percent change accounts for differences in the starting mass of the cores, so results can be compared fairly among cores.

(c) 2 points

- 1 pt — Correctly labeled axes with units (x = sucrose concentration (M); y = % change in mass), appropriate scale including negative values, and data plotted correctly. A line or scatter plot is appropriate because concentration is continuous.
- 1 pt — Error bars plotted correctly (e.g., +18.0 ± 1.6 spans 16.4 to 19.6).

(d) 2 points

- 1 pt — Estimates the isotonic concentration at about 0.32 M (accept 0.30–0.35 M): the point where percent change in mass equals zero, between 0.2 M (+6.5%) and 0.4 M (−4.2%).
- 1 pt — Calculates $\Psi_S = -iCRT = -(1)(0.32)(0.0831)(295) \approx -7.8$ bars (accept values consistent with the estimate in the first point, about −7.4 to −8.6 bars).

(e) 1 point — Predicts that the isotonic concentration would be higher for sweet potato, because its cells have a higher solute concentration (lower solute potential), so a more concentrated sucrose solution would be needed to produce no net water movement.

Question 2 (4 points)

- **(a) 1 pt** — SA = $6 \times 4^2 = 96$ µm^2; V = $4^3 = 64$ µm^3; SA:V = $96 \div 64 = 1.5$ (1.5 : 1).
- **(b) 1 pt** — The 1 µm cell has a higher SA:V (6 : 1), so it has more membrane per unit of cytoplasm for exchanging nutrients and wastes, and materials diffuse a shorter distance to reach the center.
- **(c) 1 pt** — Describes a feature and a matching cell type, e.g., microvilli on intestinal cells; root hairs on root epidermal cells; a long, thin shape in neurons; a flattened disk shape in red blood cells; folded membranes.
- **(d) 1 pt** — The mouse. It has a higher SA:V ratio, so it loses body heat to the environment faster relative to its volume and must produce more heat (higher metabolic rate) per gram to maintain body temperature.

Question 3 (4 points)

- **(a) 1 pt** — The pump uses energy from ATP to move 3 Na^+ out of the cell (and 2 K^+ in) against their concentration gradients, keeping Na^+ concentration high outside and low inside.
- **(b) 1 pt** — Identifies cotransport (secondary active transport) *and* explains that without Na^+ outside there is no Na^+ gradient to drive the cotransporter, so glucose cannot be moved against its gradient (uptake falls to 8%).
- **(c) 1 pt** — Ouabain stops the pump, so Na^+ that enters through the cotransporter (and leaks in) is no longer pumped out; the Na^+ gradient dissipates, removing the energy source for glucose transport.
- **(d) 1 pt** — At 2 minutes the existing Na^+ gradient had not yet run down; it takes time for Na^+ to accumulate inside the cell once pumping stops, so the cotransporter could still work.

INTERPRETING YOUR SCORE

Multiple choice: 22–25 correct = excellent command of Unit 2; 17–21 = solid, review missed topics; below 17 = reread the lessons for the topics you missed and redo their Check Your Understanding sets. **Free response:** 14–17 points = strong; 10–13 = good, focus on complete "because" explanations and showing calculation work; below 10 = practice the task verbs and calculations in Part 2. These ranges are study guidance only and do not predict an official AP score.

Answers: Who Am I?

Accept reasonable equivalents, such as “Golgi body” for Golgi complex or “ER” with the correct type.

SECTION A

1. Nucleus
2. Nucleolus
3. Nuclear pores
4. Ribosome
5. Rough endoplasmic reticulum
6. Smooth endoplasmic reticulum
7. Golgi complex (Golgi apparatus)
8. Vesicle
9. Lysosome
10. Autophagy
11. Apoptosis
12. Mitochondrion
13. Cristae
14. Matrix (mitochondrial matrix)
15. Chloroplast
16. Thylakoids (a stack is a granum)
17. Stroma
18. Central vacuole
19. Tonoplast
20. Peroxisome
21. Cytoskeleton
22. Microtubules
23. Microfilaments
24. Centrioles
25. Cell wall
26. Plasmodesmata
27. Nucleoid
28. Plasmid
29. Eukaryotic cell
30. Endomembrane system

SECTION B

31. Fluid mosaic model
32. Amphipathic
33. Hydrophilic (phosphate) head
34. Hydrophobic (fatty acid) tails
35. Integral protein
36. Transmembrane protein
37. Peripheral protein
38. Glycoprotein
39. Glycolipid
40. Cholesterol
41. Unsaturated fatty acid
42. Saturated fatty acid
43. Selective permeability
44. Oxygen (O_2)
45. Hydrophobic core (interior of the bilayer)
46. Aquaporin
47. Extracellular matrix (ECM)
48. Cellulose
49. Chitin
50. Peptidoglycan
51. Steroid hormone
52. Transport proteins
53. Receptor protein
54. Lateral movement (lateral diffusion)
55. Phospholipid bilayer

SECTION C

56. Concentration gradient
57. Diffusion
58. Dynamic equilibrium
59. Passive transport
60. Facilitated diffusion
61. Carrier protein
62. Gated channel (gated ion channel)
63. Saturation
64. Active transport
65. Sodium–potassium pump (Na^+/K^+ ATPase)
66. Phosphorylation
67. Membrane potential
68. Electrochemical gradient
69. Cotransport (secondary active transport)
70. Proton pump
71. Osmosis
72. Hypertonic
73. Hypotonic
74. Isotonic
75. Plasmolysis
76. Turgid
77. Water potential (Ψ)
78. Solute potential (Ψ_S)
79. Contractile vacuole
80. Osmoregulation

SECTION D

81. Exocytosis
82. Endocytosis
83. Phagocytosis
84. Pinocytosis
85. Receptor-mediated endocytosis
86. Surface area-to-volume ratio (SA:V)
87. Microvilli
88. Root hairs
89. 3 : 1
90. Compartmentalization
91. Intermembrane space
92. Endosymbiont (endosymbiotic) theory
93. Endosymbiont
94. Aerobic (oxygen-using) bacterium
95. Cyanobacteria
96. Circular
97. Binary fission
98. Infolding of the plasma membrane
99. Prokaryotic cell
100. Catalase

Glossary

Active transport
Movement of a substance against its concentration gradient, using energy, through a carrier protein.

Amphipathic
Having both hydrophilic and hydrophobic regions.

Aquaporin
Channel protein that allows rapid movement of water across a membrane.

Autophagy
Digestion of a cell's own worn-out organelles by lysosomes.

Binary fission
Prokaryotic cell division; also how mitochondria and chloroplasts divide.

Carrier protein
Transport protein that binds a substance and changes shape to move it across.

Cell wall
Rigid, porous layer outside the plasma membrane of plants, fungi, and most prokaryotes.

Central vacuole
Large water-filled sac in plant cells that maintains turgor pressure.

Centrioles
Paired microtubule structures in the centrosome of animal cells.

Channel protein
Transport protein with a hydrophilic pore.

Chloroplast
Double-membrane organelle where photosynthesis occurs.

Cholesterol
Steroid in animal membranes that buffers fluidity.

Compartmentalization
Division of the cell into membrane-bound regions with different functions.

Concentration gradient
Difference in concentration of a substance between two regions.

Contractile vacuole
Organelle that pumps excess water out of freshwater protists.

Cotransport
Coupled transport in which one substance moving down its gradient drives another up its gradient.

Cristae
Folds of the inner mitochondrial membrane.

Cytoskeleton
Network of microtubules, microfilaments, and intermediate filaments.

Cytosol
Fluid portion of the cytoplasm.

Diffusion
Net movement of particles from higher to lower concentration.

Electrochemical gradient
Combined concentration and electrical gradient for an ion.

Endocytosis
Uptake of material by inward folding of the plasma membrane.

Endomembrane system
Nuclear envelope, ER, Golgi, lysosomes, vacuoles, vesicles, and plasma membrane.

Endosymbiont theory
Theory that mitochondria and chloroplasts descend from engulfed prokaryotes.

Eukaryotic cell
Cell with a nucleus and membrane-bound organelles.

Exocytosis
Release of vesicle contents outside the cell.

Facilitated diffusion
Passive transport through a channel or carrier protein.

Fluid mosaic model
Model of the membrane as a fluid bilayer containing a mosaic of proteins and other molecules.

Glycolipid
Lipid with an attached carbohydrate chain.

Glycoprotein
Protein with an attached carbohydrate chain.

Golgi complex
Stacked membrane sacs that modify, sort, and package proteins and lipids.

Hypertonic
Having a higher solute concentration than another solution.

Hypotonic
Having a lower solute concentration than another solution.

Integral protein
Protein embedded in the lipid bilayer.

Isotonic
Having equal solute concentration.

Lysosome
Membrane sac of hydrolytic enzymes with an acidic interior.

Membrane potential
Voltage across a membrane.

Microvilli
Membrane projections that increase surface area for absorption.

Mitochondrion
Double-membrane organelle where cellular respiration produces ATP.

Nucleoid
Region containing DNA in a prokaryotic cell.

Nucleolus
Region in the nucleus where ribosomal subunits are assembled.

Nucleus
Double-membrane organelle that contains the cell's DNA.

Osmoregulation
Control of water balance and solute concentration.

Osmosis
Diffusion of water across a selectively permeable membrane.

Passive transport
Movement down a concentration gradient with no energy input.

Peripheral protein
Protein loosely attached to the membrane surface.

Peroxisome
Organelle that breaks down fatty acids and H_2O_2.

Phagocytosis
Endocytosis of large particles ("cell eating").

Pinocytosis
Nonspecific endocytosis of fluid ("cell drinking").

Plasmodesmata
Channels through plant cell walls connecting neighboring cells.

Plasmolysis
Shrinking of a plant cell's cytoplasm away from its wall.

Pressure potential (Ψ_P)
Component of water potential due to physical pressure.

Prokaryotic cell
Cell without a nucleus or membrane-bound organelles.

Proton pump
Active transporter that moves H^+ across a membrane.

Receptor-mediated endocytosis
Endocytosis of specific molecules that bind membrane receptors.

Ribosome
rRNA–protein complex that synthesizes proteins.

Rough ER
ER with bound ribosomes; makes proteins for secretion and membranes.

SA:V ratio
Surface area divided by volume.

Selective permeability
Property of allowing some substances to cross more easily than others.

Smooth ER
ER without ribosomes; makes lipids, detoxifies, stores Ca^{2+}.

Sodium–potassium pump
ATP-powered carrier moving 3 Na^+ out and 2 K^+ in.

Solute potential (Ψ_S)
Component of water potential due to solutes; $-iCRT$.

Stroma
Fluid surrounding the thylakoids in a chloroplast.

Thylakoid
Membrane sac in a chloroplast where light reactions occur.

Tonicity
Ability of a solution to cause a cell to gain or lose water.

Turgor pressure
Pressure of cell contents against the cell wall.

Vesicle
Small membrane sac that transports materials.

Water potential (Ψ)
Tendency of water to move; water moves from high to low Ψ.

Score Tracker

Record each attempt. Redo any set where you scored below 80% after reviewing the lesson.

PRACTICE SET	ATTEMPT 1	ATTEMPT 2	ATTEMPT 3	TOPICS TO REVIEW
Check Your Understanding 2.1 (/6)				
Check Your Understanding 2.2 (/6)				
Check Your Understanding 2.3 (/6)				
Check Your Understanding 2.4 (/5)				
Check Your Understanding 2.5 (/5)				
Check Your Understanding 2.6 (/5)				
Check Your Understanding 2.7 (/5)				
Check Your Understanding 2.8 (/6)				
Check Your Understanding 2.9 (/5)				
Check Your Understanding 2.10 (/4)				
Check Your Understanding 2.11 (/5)				
Practice Test — Section I (/25)				
Practice Test — Section II, FRQs 1–3 (/17)				
Who Am I? Sections A–D (/100)				

Learning Target Checklist

Check each statement only when you can do it without notes.

- ☐ I can compare prokaryotic, animal, and plant cells and explain why ribosomes are evidence of common ancestry.
- ☐ I can state the structure and function of every organelle in the Organelle Gallery.
- ☐ I can trace a secreted protein from the rough ER to the plasma membrane.
- ☐ I can calculate SA:V for a cube and a sphere and explain why cells are small.
- ☐ I can draw and label the fluid mosaic model and explain what affects fluidity.
- ☐ I can predict whether a substance will cross the bilayer and justify my prediction.
- ☐ I can compare simple diffusion, facilitated diffusion, and active transport.
- ☐ I can predict water movement using tonicity and water potential, and calculate Ψ_S.
- ☐ I can describe the sodium–potassium pump step by step and explain cotransport.
- ☐ I can compare phagocytosis, pinocytosis, receptor-mediated endocytosis, and exocytosis.
- ☐ I can explain the benefits of compartmentalization and the evidence for endosymbiosis.